

MOSAICS Studio: all-atom natural-move Monte Carlo on modern additive force fields in the browser

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Abstract

Natural-move Monte Carlo displaces a structurally meaningful group of atoms as a unit and repairs the covalent geometry with a stochastic chain closure. It has mostly been run on reduced representations under knowledge-based potentials, which return a ranking and not a quantity. We report that MOSAICS, the natural-move engine, now evaluates modern additive force fields at all-atom resolution, and that MOSAICS Studio puts that engine inside a web browser. The force fields are ff14SB and ff19SB for protein, the latter with its per-residue (ϕ, ψ) correction maps; parmbsc0, parmbsc1, OL15/OL3, OL21/OL3 and OL24/OL3 for nucleic acid; and CHARMM36 for nucleic acid, with its Urey–Bradley terms and harmonic impropers. Single-point energies were compared with sander 24.0, OpenMM 8.5.2 and GRO-MACS 2025.4 on eleven systems of 56 to 6,139 atoms, from free-ended 4-mers and capped peptides to a peptide–MHC class I complex and a protein–RNA complex, in 26 system-and-force-field combinations. Nineteen of the 26 agree with every available reference to within 0.001 kcal/mol; the median largest gap is 0.00037 kcal/mol; the worst, 0.043 kcal/mol on BPTI under ff19SB, is one improper whose atom order differs between the AMBER topology and the MOSAICS residue template. Term by term, on an 818-atom DNA:RNA heteroduplex and a capped peptide, the largest difference from OpenMM on any term is 1.8×10^{-4} and 9.6×10^{-6} kcal/mol, both on the angle term and both arising in the reference topology. The charge-to-energy constant was measured, not assumed: it differs between these programs by enough to move the heteroduplex by 0.064 kcal/mol. Analytic forces track a central difference of the same energy at second order in 61 term-and-system sweeps, and one Monte Carlo step at 818 atoms costs 3.46 ms on one core. MOSAICS Studio runs the engine, compiled to WebAssembly, and its own preparation and analysis pipeline inside the visitor’s browser tab, so nothing is installed and nothing is uploaded; a documented minimisation reproduces between the native and WebAssembly engines to the same 2,040 iterations, the same first and last energies, and 0.0045 Å between final structures, and every run carries a record naming the engine, the parameter files and the inputs by digest. The chemistry and geometries tested bound every claim: one conformation per system, in vacuum, with no cutoff.

1 Introduction

The conformational changes that matter biologically are collective: a loop folding over an occupied active site, two domains rotating against one another so that a duplex groove opens far enough to admit a protein side chain. Tens to hundreds of atoms move together, over microseconds or longer, and molecular dynamics reaches that regime slowly for a structural reason rather than an implementation one. The integrator has to resolve the fastest motion present, which is the stretch of a bond to hydrogen, so the step is a few femtoseconds and a microsecond costs of order 10^9 of them [1]; trajectories long enough to watch a domain rearrange or a small protein fold have generally required purpose-built hardware [2, 3]. The usual response is to bias the dynamics, by umbrella sampling [4], replica exchange [5], metadynamics [6] or one of their relatives [7], which buys sampling at the price of either many replicas or a collective variable that has to be guessed before the run starts.

Natural-move Monte Carlo attacks the same problem from the other side. Instead of displacing every atom by a small amount and repeating, it displaces a structurally meaningful group of atoms by a large amount, once.

The user partitions the molecule into groups (a chain, a domain, a helical arm, a backbone segment between two anchors) and nests those groups in a hierarchy. A move picks a group and applies a rigid rotation and translation to it, or turns torsions inside it, leaving everything outside where it was. That breaks the covalent geometry at the two boundaries of the group, and the break is repaired by a stochastic chain-closure algorithm that finds the alternative closed geometries, chooses among them and reports the weight needed to keep the Metropolis test exact [8]. The composite proposal is then accepted or rejected against the Boltzmann factor in the usual way [9]. The granularity is the point: one accepted move can shift a whole segment along the path a hinge would take, and the nesting lets one proposal act at the scale of a domain and the next at the scale of a sugar pucker. Nothing has to be chosen in advance about *where* the system should go, only about which parts of it are plausibly rigid. The approach has predicted sequence-dependent nucleosome occupancy without training on occupancy data [10], refined structures into single-particle EM and cryo-EM densities [11, 12], modelled epigenetic marks on DNA [13], and followed peptide dissociation from MHC class I [14].

All-atom Monte Carlo on additive force fields is not itself new, and this paper does not claim it. CAM-PARI samples all-atom polypeptide and nucleic-acid models under the ABSINTH implicit-solvent model and under imported additive parameter sets [15]; MCPRO and BOSS were built around OPLS-AA [16]; ProtoMS computes binding free energies under AMBER and OPLS parameters [17]; and Cassandra samples all-atom fixed-charge models of molecular fluids [18]. Concerted rotation closes a protein or nucleic backbone under OPLS in much the same spirit as a natural move does [19, 20]. What is new here is the combination: a nested natural-move proposal, acting at the scale of a domain on one step and a sugar pucker on the next, driven by a published additive potential whose evaluation has been checked term by term against the molecular dynamics codes that distribute it.

Two things have held the natural-move method back, and neither is the sampling. The first is the energy. Where the natural-move studies above did not use an experimental restraint they used a knowledge-based or simplified potential, on a reduced representation where the question allowed one [21], the same family of scoring functions Monte Carlo structure prediction has relied on more generally [22]. For the questions those studies asked this was the right choice: a statistical potential is cheap, it tolerates a coarse representation, and it was trained on the discrimination being requested, which is whether one model ranks above another. But ranking is all it returns. A number produced for one system cannot be compared with one produced for another, or with experiment, and a structure cannot pass between the Monte Carlo engine and a molecular dynamics run without changing what the energy means. The second is access. The engine is a command-line program driven by a plain-text input file, a residue topology file and a parameter deck, and the move set has to be written down as a hierarchy of atom groups. A PyMOL plugin, PymoSAICS [23], was written because defining those groups by hand was the step users found hardest, and it remains a plugin: it lives inside PyMOL and reaches only the users who already have the engine. Closing both gaps is what this paper is about.

The energies to adopt are the additive force fields molecular dynamics already uses, and the specific ones matter, because each was written to repair a named failure of its predecessor. For protein, ff14SB refits the backbone ϕ/ψ and side-chain χ dihedral terms of ff99SB against quantum reference data [24]; ff19SB then removes the backbone dihedral terms altogether and replaces them with amino-acid-specific two-dimensional (ϕ, ψ) correction maps fitted to solution-phase surfaces [25], using the grid-based CMAP cross-term introduced for CHARMM a decade and a half earlier [26, 27]. For nucleic acids, parmbsc0 refits α and γ and so removes the irreversible backbone flips that used to destroy long DNA trajectories [28]; parmbsc1 goes further into ϵ/ζ , χ and sugar pucker, benchmarked against a large set of experimental DNA structures [29]. The Olomouc corrections take a different route: χ_{OL3} for RNA [30], a β refit that gives OL15 for DNA [31], an α/γ refit against Z-DNA that gives OL21 [32], and a deoxyribose pucker refit that gives OL24 [33]. The DNA variants among these have been compared against experiment over long trajectories [34, 35]. CHARMM36 is a separately developed alternative with a different functional form, carrying Urey–Bradley 1–3 terms and harmonic impropers, refit for the DNA BI/BII equilibrium [36] and for RNA 2'-hydroxyl sampling [37].

These parameter sets are not interchangeable, and they differ from one another substantially. How much they differ is not something that can be read off an absolute energy at a single geometry, and this paper does not try: a Fourier torsion sum carries an arbitrary additive constant, so a refit moves the constant as freely as it moves the shape, and the difference between two parameter sets' absolute torsion energies at one

conformation is not a physical quantity. What an all-atom engine buys is narrower and more useful. The number it returns is the published function's, evaluated on a named parameter set, so a preference the sampler reports can be attributed to that parameter set, and an ensemble it produces and an ensemble a molecular dynamics code produces under the same parameters are ensembles of the same function. A knowledge-based potential returns a ranking on a scale of its own and can do neither.

This paper reports two things. The first is that MOSAICS evaluates these force fields at all-atom resolution, and how well. Little of that is new engine code. MOSAICS reads a force field as a deck of plain-text parameter files, and the code that evaluates those terms was already there. CHARMM36's Urey-Bradley 1-3 terms go into a slot the bend record already carried, evaluated by the expression the bend routine already applied to it; its 188 harmonic impropers go onto the power series the torsion record already expanded. The correction-map machinery was present and its energy was already correct, but its analytic force was identically zero at a dihedral of exactly π , which is where a peptide built extended sits. That repair, a memory-allocation defect that ran on every committed job, and a deck reader that carries a per-molecule label on torsion entries and chains the residues it builds are the changes to engine code reported here; the released engine cannot read most of these decks at all. The rest of the work is conversion: the decks, the converters that write them, and the defects that writing them exposed. The verification is term by term, on the same coordinates, in vacuum and with no cutoff, at one fixed conformation per system, against three independently written programs, sander 24.0, OpenMM 8.5.2 on the Reference platform and GROMACS 2025.4 in double precision [38, 39, 40, 41], following the cross-engine practice established for this purpose [42, 43]. It covers eleven systems of 56 to 6,139 atoms and 26 system-and-force-field combinations (Figure 1), and it is completed by a finite-difference check of the analytic forces and by a measurement of what one Monte Carlo step costs.

The second is MOSAICS Studio, which puts that engine in front of a user who has never seen its input syntax, in a web browser. The engine is compiled to WebAssembly and the Studio's own pipeline, four operations, sequence to structure, structure to minimised structure, variant building and structure comparison, runs beside it under Pyodide, so a sequence or a coordinate file goes in, a minimised structure and a molecular graph come out, nothing is installed and nothing leaves the visitor's machine. Every run carries a record naming the engine build, the parameter files and the inputs by digest. The WebAssembly engine was checked against the native engine on a documented minimisation, and that number is reported alongside the force-field ones, because the Studio is only as good as the engine it carries.

After every electrostatic term is put on a single Coulomb constant, nineteen of the 26 combinations agree with every reference available to them to within 0.001 kcal/mol, the median largest gap over the 26 is 0.00037 kcal/mol, and the one combination that is worse than 0.01 kcal/mol is BPTI under ff19SB, where a single improper is read in a different atom order by the AMBER topology and the MOSAICS residue template. On the two systems compared term by term, the largest difference between MOSAICS and OpenMM on any term is 1.8×10^{-4} kcal/mol on the 818-atom DNA:RNA heteroduplex and 9.6×10^{-6} kcal/mol on the capped peptide, in both cases on the angle term, and in both cases traceable to the reference topology and not to MOSAICS. Analytic forces were checked against a central difference of the same energy in 61 term-and-system combinations; fitted over a fixed window, the convergence exponent lies between 1.994 and 2.009. One Monte Carlo step at 818 atoms costs 3.46 ms on one core of a laptop.

Section 2 describes the systems, the parameter decks, how each reference calculation was produced, and how the Studio is built and served. Section 3 gives the 26-cell comparison, the term-by-term comparison for the two reference systems, the electrostatic-constant correction, the force checks, the cost measurement and the Studio's own verification. Section 4 sets out what follows from single-point agreement, and what does not.

2 Methods

Everything reported in Sections 3.1 to 3.5 is a single-point energy or a force at one fixed geometry. No structure moves during any of those runs. The protocol removes every difference between the programs being compared except the potential itself, so that a disagreement can only come from the parameters or from the code that evaluates them. What a reader needs only in order to rerun a job, the engine and reference control

files, the deck sizes, the `t1eap` presets, the CHARMM36 route and the `1efs` coordinate edits, is tabulated in the Supporting Information and appears here only where a choice carries an argument.

2.1 The engine and how it was run

MOSAICS [44] is the natural-move Monte Carlo engine of Sim *et al.*[9] and Minary and Levitt[8]. It was built from the serial target, `source/compile/serial`, of the engine’s main development branch and run from the command line. Every run directory in the evidence tree records the SHA-256 of the binary that produced it and the commit it was built from, and that is how a build is identified, because the banner the engine prints reads `version.3.9.1` on every build used here. The released 3.9.1 binary was compared against the development build on the same inputs before the comparison began: it refuses the OL21/OL3, `parmbsc1` and `parmbsc0` decks at the deck, because their torsion entries carry a per-molecule label it does not know; it refuses the `ff19SB` input because of the correction-map term, and with that term stripped it omits the term silently; and on the OL15/OL3, `ff14SB` and CHARMM36 decks it runs and returns energies tens of thousands of kcal/mol from the references, because it does not chain the residues it builds from a deck. The development build reproduces every committed single point byte for byte. The gate report is in the evidence tree. There is no software paper for the engine, so it is cited by name and project page.

A MOSAICS job reads four things: a simulation input file, a residue topology file (RTF) naming the atoms of each residue and their types, a set of parameter files (a *deck*), and a PDB structure. Supporting Information Table S1 lists every setting that makes a run a single point. Four of them decide what is measured. Every proposal width is zero, so the one Monte Carlo step taken proposes the starting structure. `\energy_report{2}` makes the engine print each term and each part of each term, and `\inter_list{none}` switches the neighbour list off, with no interaction cut off or switched. The nucleic runs use torsional proposals (`prop_type tors`) with a `time_step_md` of 0.4, the peptide runs Cartesian proposals (`prop_type cart`) with a time step of zero: in Cartesian mode a non-zero step lets the structure drift even at zero proposal width, and the energies then describe two different shapes. The fourteen committed inputs of the two reference systems and the two controls also run clean under the address sanitizer with `alloc_dealloc_mismatch` turned on, which is off by default on macOS.

What is new in the engine, and what is not. Most of this work is conversion and reference construction, not engine code. The six energy terms (bond, bend, torsion and improper, one-four, non-bonded, correction map), their analytic forces and the staging transform all predate it. Three things changed in engine source. The deck reader carries a per-molecule label on torsion entries and chains the residues it builds, which is what the released 3.9.1 lacks. The correction-map analytic force was identically zero at a dihedral of exactly 0 or π , repaired in `cmap.cpp` (finding F-21); the correction-map *energy* was never affected. And forty arrays allocated with `new[]` and grown with `realloc` were made consistent across ten source files (F-19). The two CHARMM terms AMBER does not have needed no engine change at all, riding on records the engine already evaluates. Finding identifiers throughout follow `evidence/findings.txt`, the register in which an identifier is never reused.

The path a sampling run takes. No run in Sections 3.1 to 3.5 samples, so it is worth saying which code a sampling run would execute. MOSAICS scores a proposal by recomputing the whole potential: `Pos::potential`, in `source/energy/energy.cpp`, calls the bond, bend, torsion, one-four, non-bonded, correction-map and constraint routines over the entire molecule on the trial coordinates, and the Metropolis ratio is formed from two such full potentials. There is no incremental ΔE over a displaced subset, so the term routines validated here are the term routines a proposal is judged by. One difference remains. `\inter_list` is `none` in every single-point run, so the non-bonded sum runs over all non-excluded pairs, while a production run may select the link-list or neighbour-list branch; those branches restrict the pair set of the same expression and are not exercised by any single point here. The cost of a proposal is one full evaluation, quadratic in the number of non-excluded pairs, and Section 2.8 measures it.

2.2 Force fields, and how a parameter set enters the engine

MOSAICS reads a force field as a deck of five plain-text files alongside an RTF: `.bond`, `.bend`, `.tors_and_impr`, `.onfo` for atoms three bonds apart and `.vdw` for everything further. The decks used here are `db_ff14sb` and `db_ff19sb` for protein; `db_bsc0`, `db_bsc1`, `db_ol15_ol3`, `db_ol21_ol3` and `db_ol24_ol3` for nucleic acid, the last three each with a `_terminal` variant; `db_charmm36_na`; and, for the protein–RNA complex, two decks that hold protein and RNA types together, `db_ff14sb_ol3` and `db_ff19sb_ol3`. Supporting Information Table S2 collects the entry counts the engine prints as it reads the six decks of the two reference systems. How many of those entries a given test system actually evaluates is not reported by the engine and is not reported here.

The protein sets are `ff14SB`[24] and `ff19SB`[25]. `ff19SB` replaces the `ff14SB` backbone dihedral terms with per-residue two-dimensional (ϕ, ψ) correction maps, written in the grid-based CMAP form introduced for CHARMM[26, 27], so its deck carries a sixth file, `.cmap`, and its input carries `\energy_term{cmap}` and `\cmap_database_file`. On the capped tetrapeptide the engine reads 44 database entries, loads 16 maps ($N = 24$) and selects four of those sixteen, one per interior residue.

The nucleic sets are `parmbsc0`[28], `parmbsc1`[29], `OL15`[31], `OL21`[32] and `OL24`[33] for DNA, each paired with χ_{OL3} [30] for RNA. Every torsion entry carries a per-molecule label. The two `bsc` decks originally carried none, and without it the four atom-type quadruplets that need different values for DNA and for RNA collapse onto the DNA values, which is worth 0.47 kcal/mol on the 1e5 torsion energy (F-03). The one-four electrostatic scale factor is stored as exactly 5/6 in every deck; storing it to seven or ten digits instead shifts the 1e5 total by 1.0×10^{-4} kcal/mol (F-04).

Each deck is converted from a published parameter file and named after the source it came from. The `OL15/OL3`, `OL21/OL3` and `ff14SB` decks were written by `tools/openmm_xml_to_mosaics.py` from the OpenMM force-field XML files distributed with OpenMM and `openmmforcefields`, overlaid with the exact proper-torsion rows of a matched AMBER `tLeap` topology so that wildcard and exact torsion precedence match `sander`. The `OL24/OL3` decks were written by `tools/amber_prmtop_to_mosaics.py` from resolved AMBER topologies, the DNA half from the `ff-DNA-OL24` package distributed by the `OL24` authors and the RNA half from the `OL3` files in `AmberTools`. Regeneration was checked with the unmodified converter as a control: `db_ff14sb` regenerates byte-identical; `db_ol15_ol3` does not regenerate at all, and neither does the `tors_and_impr` file of `db_ol21_ol3` (F-20), so what backs those two decks’ provenance is a note in the repository rather than a command a reader can run. The `OL15/OL3` and `OL21/OL3` rows should be weighed accordingly.

Standard and terminal decks. A standard nucleic deck expects a phosphate on every residue, which is what the padded heteroduplex build provides. A chain with a real 5′-hydroxyl end selects different rows, so `OL15`, `OL21` and `OL24` each carry a `_terminal` deck written from true-terminal tetranucleotide topologies, and it is the terminal deck that every free-ended system in this work reads. `parmbsc1` has no terminal deck. Run on the Drew–Dickerson dodecamer through its standard deck it returns a torsion energy 0.385 kcal/mol above the references, and the same experiment with `OL21/OL3`’s standard deck in place of its terminal one reproduces the size of that error (0.369 kcal/mol), so the gap is the missing terminal profile and not the parameters. `parmbsc0` does not run on that structure at all. Neither combination is in the 26-cell matrix, and both are recorded as open findings in the evidence tree.

CHARMM36. The CHARMM36 nucleic acid parameters[36, 37] reach the engine by a different route, recorded in `evidence/charmm36.txt` and set out step by step in Supporting Information section S6. Deck and reference energies are both read out of an OpenMM System built from one release, `charmm36_2024.xml`, so the comparison does not straddle two releases, but the numbers belong to that one and no release-independent claim about CHARMM36 is made here. The two CHARMM terms AMBER does not have ride on records the engine already has, the Urey–Bradley 1–3 harmonic on `~bend_parm` and the 188 harmonic impropers on `~torsion_parm`, where the deck reader expands them into a truncated sixth-order Fourier series. That series is the conversion’s only genuine approximation: exact to four figures at 10° from equilibrium and −7.7% at 60°, with the intermediate angles in Table S6, and nucleobase impropers are stiff enough to stay inside 20°. CHARMM36 here is nucleic acid only, only on the 5′-hydroxyl terminus, and

Table 1. The engines, the resolution of the file each number is read from, the floor that resolution puts under a difference, and the committed file driving each run. All four run in double precision. D is the directory for one force field and system; “d.p.” is decimal places as printed. A composite term sums two printed columns, so its floor is twice the per-column figure, about 2×10^{-4} for sander and 4.8×10^{-7} for GROMACS.

Program	Version	Output resolution	Floor (kcal/mol)	Input
MOSAICS	main branch, serial	16 significant figures	not limiting	D/mosaics/*.input
sander	24.0 (AmberTools 24.8)[38]	4 d.p., kcal/mol	10^{-4}	D/sander/sander_sp.in
OpenMM	8.5.2[39, 40]	12 d.p., kcal/mol	5×10^{-13}	openmm_single_point.py
GROMACS	2025.4[41]	6 d.p., kJ/mol	2.4×10^{-7}	D/gromacs/sp.mdp

only against OpenMM; the protein force field[64] is not implemented, because two of its correction-map selectors distinguish a proline neighbour by the class of the *following* residue while ~cmap_binding selects a map by the residue itself, so supporting it changes the file format and the engine (F-09).

2.3 Reference engines

Three independently written programs serve as references (Table 1), and where they agree with each other the target is not in doubt. No number here is produced by a program that reads another program’s output; each writes its own output file and those files are committed. Term-by-term cross-engine comparison of this kind has an established precedent[42], and the unit and convention traps it has to avoid are catalogued in ref. [43].

Those floors are properties of the reference and not of MOSAICS, each a half-unit in the last place that reference prints. OpenMM’s is the one that invites a misreading: it prints twelve decimal places in kcal/mol, not twelve significant figures, so its floor is an absolute quantity and does not scale with the size of the term. The rule this paper follows is that no difference smaller than the coarser of the two engines’ floors is reported as a measurement; below that floor a difference is reported as below the resolution of the reference output, with the floor stated, and is not read as agreement to that many digits.

The three control files are reproduced option by option in Supporting Information Table S3. sander is run as a zero-cycle minimisation in vacuum with no cutoff at dielectric 1, and its energies are read from the FINAL RESULTS block. OpenMM is driven on the same topology and coordinate files sander reads, on the Reference platform in double precision with no cutoff, no constraints and no centre-of-mass removal. It reports one combined non-bonded energy, so the system is evaluated twice more, once with every charge zeroed and once with every Lennard-Jones ϵ zeroed, to split it; the two halves and the combined value are all printed, and their difference is a self-check the reader can confirm at 1.5×10^{-11} kcal/mol on 1efs. GROMACS is run as a rerun over the single frame with plain cut-off electrostatics and Lennard-Jones, no modifiers, no dispersion correction, no constraints, and the cutoff set past the far end of the molecule so that every pair is inside it with 2 nm to spare and no periodic image is; sp.log confirms a potential shift of zero on every term. Energies are read from sp.xvg, never from what gmx energy prints on screen, which is rounded coarsely enough to look like a disagreement between engines. GROMACS references exist for the six original AMBER combinations and for the ff14SB proteins and the OL21/OL3, OL15/OL3 and OL3 nucleic systems; the ff19SB rows carry no GROMACS total, for the reason given next.

The GROMACS conversion route, and what it does not carry. The GROMACS topology is produced mechanically from the same AMBER topology the other two engines read, by the ParmEd command in evidence/engines.txt; ParmEd is used as distributed with AmberTools[38]. In the topology this route produces, [cmaptypes] is keyed on the five backbone atom types, which every residue of a protein shares, so ff19SB’s per-residue maps collapse onto one (F-06). On the four-residue peptide that costs 0.30 kcal/mol; on ubiquitin the GROMACS correction-map energy is 61.54 kcal/mol against the 69.95 that MOSAICS and OpenMM both return. That is a limitation of the conversion route as used here, not of the GROMACS file format: published GROMACS ports of amino-acid-specific CMAP force fields introduce residue-specific backbone atom types so that each map gets its own type quintuple. That workaround was not attempted, so ff19SB is reported as verified against sander and OpenMM, and the GROMACS column of every ff19SB row is blank rather than wrong.

Table 2. The eleven test systems. Atom counts are read from the built topology, hydrogens included; the force fields are those with at least one reference engine in the 26-cell matrix of Table 3. All are run in vacuum at dielectric 1 with no cutoff. Source: `work/preprint-I/systems/MANIFEST.json`, which also records the SHA-256 of every coordinate file.

System	Class	Atoms	Chains	Force fields compared	What it tests
1efs	heteroduplex	818	2	OL21/OL3, OL15/OL3, OL24/OL3, CHARMM36	DNA:RNA hybrid, model 1 of PDB 1EFS; carries the term-by- term comparison
1BNA	DNA duplex	758	2	OL21, OL15, CHARMM36, OL24	Drew–Dickerson dodecamer d(CGCGAATTCGCG), B-DNA duplex
d(ACGT)	terminal control	126	1	OL21, OL15, OL24	DNA 4-mer, one of each base, free 5'-OH and 3'-OH
2KOC	RNA hairpin	447	1	OL3, CHARMM36	UUCG tetraloop RNA hairpin, model 1 of PDB 2KOC
r(ACGU)	terminal control	127	1	OL3	RNA 4-mer, one of each base, free 5'-OH and 3'-OH
1UBQ	protein, globular	1,231	1	ff14SB, ff19SB	Ubiquitin, β -grasp fold
5PTI	protein, globular	892	1	ff14SB, ff19SB	BPTI, three disulfide bonds
ACE-AGSV-NME	protein, peptide	56	1	ff14SB, ff19SB	Capped tetrapeptide, four residue kinds, built extended
Tax 11–19	protein, peptide	165	1	ff14SB, ff19SB	HTLV-1 Tax nonamer LLF- GYPVYV, capped, free in solution
1HHK	protein–peptide complex	6,139	3	ff14SB, ff19SB	HLA-A*02:01, β_2 -microglobulin and the bound Tax nonamer
1F7Y	protein–nucleic complex	3,321	2	ff14SB/OL3, ff19SB/OL3	Ribosomal protein S15 bound to a 16S rRNA fragment

The engines do not group their terms identically, so all comparisons use three composite terms. *Electrostatics* means all of it: EEL + 1–4 EEL in sander, Coulomb-14 + Coulomb (SR) in GROMACS, the epsilons-zeroed evaluation in OpenMM, Total Coulomb energy in MOSAICS; *Lennard-Jones* is the matching sum, and *dihedral* includes impropers in all four programs.

2.4 Test systems

Eleven structures are used (Table 2, Figure 1). Two of them, the 818-atom DNA:RNA heteroduplex and the capped tetrapeptide, carry the term-by-term comparison against all three references; the other nine widen the chemistry the force fields are exercised on, from four nucleic bases to eight, from four amino acids to twenty, and from interior residues to real chain ends, disulfides, a bound peptide and a protein–RNA interface. A single-point comparison is defined by its coordinates, so each file’s provenance is given here.

1efs. A heteroduplex of one DNA strand and one RNA strand, model 1 of Protein Data Bank entry 1EFS[45, 46]. Chain A is DG DA DG DA DG DG DA DA DG DA DG DA DA, six DG at 33 atoms each and seven DA at 32, giving 422 atoms; chain B is RU RU RC RU RC RU RU RC RC RU RC RU RC, seven RU at 30 atoms and six RC at 31, giving 396. Each residue carries a charge of -1 , so the net charge is -26 and the run uses `\neutralize{off}`. It is the only structure here that mixes DNA and RNA in one duplex, and that is why it carries the term-by-term comparison: a parameter set correct for DNA and wrong for RNA is invisible on a pure duplex of either, and this is the system on which F-03 became visible. The coordinates are taken as deposited apart from six built atoms, a 5' phosphate on each strand, and four renaming or renumbering edits, all listed in Supporting Information section S7. The hydrogens are the deposited NMR hydrogens; none was rebuilt, and nothing was minimised. All four AMBER presets build the same padded system of 942 atoms in 30 residues, then trim four terminal residues to the 818 atoms compared, so every residue that enters the term-by-term comparison is an interior one. The CHARMM36 cell on this system is a different molecule: the distributed XML has no patch that both keeps a 5' phosphate and ends a chain (F-08), so the structure was rebuilt as a 5'-OH/3'-OH strand, 816 atoms, six phosphate atoms removed and four hydroxyl hydrogens

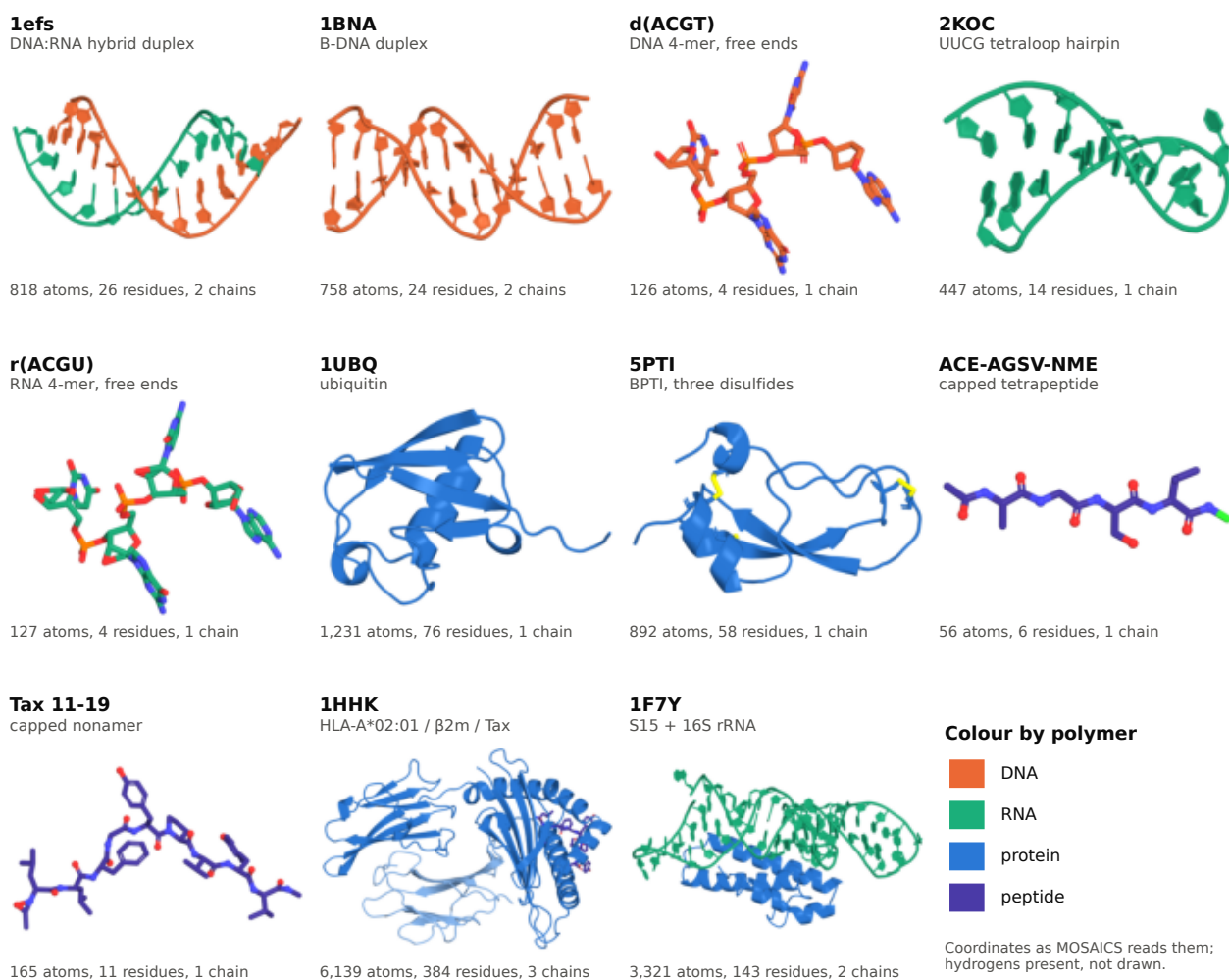


Figure 1. The eleven systems, drawn from the coordinates the engine reads. Nucleic acids and proteins are shown as cartoons, the peptides and the two free-ended 4-mers as sticks with heteroatoms coloured by element; hydrogens are present in every file and omitted from the drawing. Chain colour follows the polymer, with the second protein chain of 1HHK (β_2 -microglobulin) in a lighter blue and its bound Tax nonamer as violet sticks.

added, and its total is not comparable with the AMBER rows.

Capped tetrapeptide. ACE-ALA-GLY-SER-VAL-NME, 56 atoms, with four residues of four different kinds between the two caps. The four kinds are deliberate: each interior residue produces one (ϕ, ψ) correction term, ff19SB carries a different map for different residues, and a peptide with one interior residue would resolve its single term through a fallback and return the same energy whether the map selection were right or wrong. The coordinates come from `t1eap` and are not minimised: what `t1eap` writes is an extended chain at $\phi = \psi = 180^\circ$ on every residue. Two consequences run through the paper. The out-of-plane terms sit exactly at their minima, so both MOSAICS and GROMACS report zero for them and the comparison on that term is vacuous (F-11); and all four correction-map terms sit at $\phi = \psi = \pi$, the singular point of the implemented force expression, which is why every energy comparison passed while the analytic CMAP force was identically zero (F-21).

The free-ended 4-mers. d(ACGT), 126 atoms, and r(ACGU), 127 atoms, one of each base with free 5'-OH and 3'-OH ends. They isolate the terminal residues, which is where the residue templates of two force fields differ, and they carry the force sweeps, the CHARMM36 comparison and, through the terminal decks, AMBER comparisons against sander and OpenMM. d(ACGT) carries a geometric defect at residue DG 3, finding F-17, set out with its consequences in Section 4.6.

The deposited structures. 1BNA, 1UBQ, 5PTI and 2KOC were fetched from the RCSB on 25 August 2026 and prepared under one set of rules, recorded in `evidence-v2/_structures/BUILD-DECISIONS.md`. Only ATOM records survive; every water, ion and ligand is dropped. Alternate location A is kept. Every hydrogen and deuterium in the deposit is stripped and rebuilt by the topology builder, so hydrogen placement is `t1eap`'s for the AMBER decks and the OpenMM template's for the CHARMM path, never the depositor's. Protein termini are charged, NH_3^+ and COO^- , because a whole protein is not a fragment; nucleic termini are 5'-OH and 3'-OH. 1BNA is the Drew-Dickerson dodecamer d(CGCGAATTCGCG)[47], both strands kept because the duplex is self-complementary and one strand is not the dodecamer, 758 atoms, net charge -22; its terminal residues build as DC5 and DG3. 2KOC is model 1 of the 20-model NMR ensemble of the 14-mer cUUCGg tetraloop hairpin[48], GGCACUUCGGUGCC, 447 atoms; its deposited 5'-terminal phosphate is removed so that the chain starts at a 5'-OH, because a bare 5' phosphate is the one terminus the CHARMM36 XML cannot end a chain on, and keeping it would have bought a second system CHARMM36 cannot run. 1UBQ is ubiquitin[49], chain A, 76 residues, 1,231 atoms, net charge 0; its one histidine, HIS 68, is built as HIE, and the C-terminal tail deposited at occupancy 0.25-0.45 is taken as deposited. 5PTI is bovine pancreatic trypsin inhibitor from the joint X-ray and neutron refinement[50], chain A, 58 residues, 892 atoms, net charge +6; its three disulfides, CYS 5-55, 14-38 and 30-51, are built as CYX and bonded in `t1eap` exactly as the deposited SSBOND records give them, and the 103 deuterium positions of the neutron model are stripped with the hydrogens.

The peptide and the two complexes. The Tax nonamer LLFGYPVYV, the HTLV-1 epitope presented by HLA-A*02:01, is taken from the engine's own example of that peptide, capped with ACE and NME, 165 atoms, at the main-branch commit the run record names. 1HHK is the crystal structure of HLA-A*02:01 with β_2 -microglobulin and that nonamer[51], three chains of 275, 100 and 9 residues, built with `t1eap` under ff14SB and ff19SB from the prepared coordinates, 6,139 atoms with 2,978 hydrogens added, net charge -9. 1F7Y is ribosomal protein S15 bound to a fragment of 16S rRNA[52], chain A an 86-residue protein with charged termini and chain B a 57-nucleotide RNA, 3,321 atoms, net charge -49. It is the first system run against a deck that holds protein and RNA atom types together, and those two decks live on an engine branch that has not been merged, which Table 3 records. Its reference coordinates were written from the three-decimal PDB the engine reads rather than from `t1eap`'s restart file, because the restart differs from the PDB by up to 5×10^{-4} Å, which is worth 0.66 kcal/mol of Lennard-Jones energy on this system.

One rule governs every build. *The structure MOSAICS reads is the structure the reference topology is built from, never the reverse.* MOSAICS reads the PDB and the reference engines read the topology, and the two formats store positions to different numbers of decimals; the largest resulting difference on the heteroduplex was 0.0005 Å. On the peptide that moved the short-range energy by 0.573 kcal/mol, because that term rises as the twelfth power of distance, and it looked exactly like a broken force field (F-02). The builders write the

reference positions from the structure MOSAICS reads and refuse to continue if the two disagree.

2.5 Run conditions

Every single-point run is in vacuum, at dielectric 1, with no cutoff, no temperature coupling, no solvent and no move. No temperature enters any number reported in Sections 3.1 to 3.5: nothing is propagated, and the one proposal that is made reproduces the current structure, so the energy difference it is judged on is zero. A comparison counts only if the structure did not move, and the runs check that. MOSAICS prints each part of a term next to that term's total, so the parts must add up. On 1efs with OL21/OL3 they do so to within 5.5×10^{-11} kcal/mol, the summation order of a double-precision sum over 818 atoms; a larger discrepancy means the structure moved.

2.6 The electrostatic constant

The four programs do not share the constant that converts charge into energy, and on 1efs the disagreement is worth more than everything else being measured. It was measured, not looked up. One AMBER topology holding two point charges, $+1 e$ and $-1 e$, exactly $1.0000000000 \text{ \AA}$ apart, with Lennard-Jones zeroed, in vacuum and with no cutoff, is handed to every engine; the electrostatic energy an engine then prints is its constant. The script is `measure_coulomb_constant.py` in `evidence/coulomb-constants/`, and the four values it returns are in Table 7. sander's is measured to four digits because it prints four decimals; its exact value is $18.2223^2 = 332.052217$, because AMBER stores topology charges pre-multiplied by 18.2223. MOSAICS's row is analytic rather than measured: it is the product of two constants compiled into the engine, $0.529177 \times 627.50921 = 332.06344122017$, and that unrounded product is the number every correction is done with.

Before any comparison, the electrostatic term of the MOSAICS and sander columns is scaled by the ratio of its own constant to a single target. That target is not any engine's measured value but a number derived from the 2019 SI values of e and N_A , the thermochemical calorie and the CODATA 2018 ϵ_0 , given in Section 3.4: $k = 332.0637133 \text{ kcal \AA mol}^{-1} e^{-2}$. The OpenMM and GROMACS columns are left as printed: both print six decimals and to those six decimals both equal the derived target. Every total in the 26-cell matrix and every electrostatic residual in the term-by-term tables is on this constant.

2.7 Force validation

The analytic force is compared with a central difference of the same energy, atom by atom and component by component, for each term separately and for the total. The step is swept over ten values from 10^{-2} to 3×10^{-7} Bohr, and what is recorded at each step is the largest $|\text{numerical} - \text{analytic}|$ over every atom and component, in Hartree per Bohr. Sixty-one such sweeps are committed in `evidence/forces/`: thirty-seven against Cartesian coordinates on 1efs, the two 4-mers and the peptide, and twenty-four against a staging coordinate, the fourth staging system being a capped peptide fixture whose NME cap is not a physical amide and whose energy is therefore recorded, not validated (F-18).

The measurement is the slope, not the residual. A central difference is wrong at order step^2 , so a force that is the derivative of the energy makes the residual fall as the square of the step until round-off in the energy takes over, while a force wrong by a fixed amount stops falling at that amount. The slope is fitted by least squares on $\log_{10}(\text{residual})$ against $\log_{10}(\text{step})$ over a window declared in advance, 10^{-2} to 3×10^{-4} Bohr, four of the ten steps and spanned by every sweep. Fixing the window matters, because a fit over rows selected for already falling as the square would test nothing.

2.8 The cost of one Monte Carlo step

A single run is setup, parse, one initial energy, N steps and output. At $N = 1$ the constant part is 99.8 % of the wall clock, so timing one run and calling it the cost of one evaluation would overstate it by more than two orders of magnitude. The constant was measured instead of assumed. The committed 1efs OL21/OL3 input was run unmodified except for `\total_step_mc` at $N = 0, 1, 5, 10, 25, 50$ and 100, five repeats each, and

$t(N) = a + bN$ was fitted by least squares on the medians; b is the marginal cost of a step and a is everything that happens once. Medians rather than means because this is a laptop: the $N = 0$ point has one 866 ms outlier against a 442 ms floor. The build was the serial target at commit `eccdf74`, Apple clang 21.0.0 at `-03`, one process on one core of an Apple M5, and the timed run is the committed run: its initial total energy, 2526.184857387672 kcal/mol, is byte-identical to the committed single-point output. Acceptance over the timed steps was 59 of 100. A rejected step still costs a full energy evaluation, since the evaluation is what decides it, so the slope is an average over a stable accept/reject mix and not a cost conditional on acceptance.

2.9 MOSAICS Studio

MOSAICS Studio is a web page. Behind it is a Python package (Python 3.10 or later, MIT licence) whose pipeline is headless: four functions carry every operation, `sequence_to_structure`, `structure_to_minimised`, `build_variant` and `compare_structures`, and the page calls those functions through Pyodide. The engine is the MOSAICS executable compiled to WebAssembly and driven through a small JavaScript bridge: the pipeline writes the input file, the residue topology, the deck and the structure into a run folder on the in-tab file system, runs the engine on it, follows its progress from the lines it prints and reads back every file it writes.

The guided workflow. The page presents four steps over that pipeline. *Start* takes a sequence (DNA, RNA or a peptide, a pasted or dropped FASTA record) or a coordinate file, and builds nucleic acids on ideal A- or B-form geometry as single strands, duplexes or DNA-RNA hybrids; peptides are built on an ideal backbone with side chains and pH-7 protonation from PDB2PQR's residue definitions[54, 55] and optional ACE/NME caps; proteins are prepared from coordinates, not built. *Prepare* chooses the chains to keep, leaves ligands and waters out, adds hydrogens to a deposited protein, and chooses the force field from every compatible profile with the reason the default was chosen; thirteen profiles are declared, twelve over the AMBER and CHARMM36 decks of Section 2.2 and one knowledge-based three-point protein model, and each names its six files. One validation gate sits on this step: whether the ordinary bonded and non-bonded energies for every residue type present reproduce a committed reference. *Run* offers fifteen protocols, each with the deck it comes from, the work it is cited to and a link to the engine's manual, and exposes every parameter the engine reads for that kind of calculation, grouped as the manual groups them; the deck the engine will read is one click away. *Understand* shows the energy along the run, the RMSD to the starting structure along the trajectory, energy against RMSD with one point per frame, acceptance tallies, the energy table, a residue-by-residue comparison of before and after, the molecular graph export (NumPy, JSON, PyTorch Geometric and DGL formats [56, 57], with no learned embeddings), and the record. The three-dimensional view is 3Dmol.js[58], and a viewer plays the trajectory frame by frame.

The record. Every run writes a record naming the Studio version, the engine, the SHA-256 of the engine executable, the SHA-256 of each force-field file that was staged, the preset and every parameter it resolved, the inputs and the outputs by digest, and the start and end times. The record and every file it names can be downloaded from the page, and the package's verify routine recomputes every digest and reports whether the recorded files are present and unchanged. The engine itself already writes every parameter it resolved to `sim_param.out`; what the record adds is the binding of those parameters to the engine build, the deck files and the inputs in one document that can be re-checked.

The engine in the browser. The engine was compiled to WebAssembly with Emscripten 6.0.9[59] from the serial target of the engine's main branch at commit `73f598a` (7 September 2026). The 151 objects compile with no source changes; the link needs a 32 MB stack, because the input parser overruns Emscripten's 64 KB default at once, and one guard on the single `popen("date")` call, made in the scratch copy used for the build and not in the engine repository. The result, `mosaics.wasm`, is 713,184 bytes. The Studio's own pipeline runs beside it under Pyodide 0.27.7[60] in a Web Worker; one Python shim registers the WebAssembly build as the engine and replaces the one function that would have launched a native executable, and everything else is the package's own code. The distribution is about 8.5 MB of static files (the page, the engine, the Python archive, one archive per force-field profile fetched only when that profile is chosen, the manuals and a

version manifest); a first visit downloads about 25 MB including Pyodide and NumPy from a content delivery network, cached afterwards. Every URL the page fetches carries the build identifier, so a redeploy is never served from a browser’s cache. Nothing is uploaded and no server computes: any static host serves it, and the copy reported here is on a personal web page at the University of Oxford. The engine is single-threaded in the browser, a run cannot be interrupted from outside, and everything lives in the tab’s memory until it is downloaded.

Tests. The package carries 84 test files with 2,198 test functions (counted on 8 September 2026), run on every push on three operating systems across Python 3.10 to 3.13, and the browser build is checked end to end by a headless test that drives the whole pipeline under Node with the same Python archive and the same bridge the page uses, and fails unless the documented case study reproduces (Section 3.7).

3 Results

3.1 What was implemented and what was tested

Eight parameter families were converted into MOSAICS decks and evaluated at all-atom resolution: ff14SB [24] and ff19SB [25] for protein; parmbsc0 [28], parmbsc1 [29], OL15/OL3 [31, 30], OL21/OL3 [32, 30] and OL24/OL3 [33, 30] for nucleic acid; and CHARMM36 for nucleic acid [36, 37]. Across the eleven systems they give the 26 system-and-force-field combinations of Table 3, 54 MOSAICS-to-reference comparisons in all, 20 of the 26 cells carrying two or more references.

The cells were not all checked to the same strength, and the table says which is which. The six original AMBER combinations on the heteroduplex and the tetrapeptide were compared term by term against all three references, and the ff14SB proteins and the OL21/OL3, OL15/OL3 and OL3 nucleic systems have a GROMACS total as well as sander and OpenMM. Every ff19SB row is against sander and OpenMM only, because the ParmEd route collapses its per-residue correction maps onto one (F-06, Section 2.3); the GROMACS cell is blank, not failing. The CHARMM36 and OL24 rows are against OpenMM only. CHARMM36 has one reference because its deck and its reference are both read out of the same OpenMM build; on the Drew–Dickerson dodecamer the sander OL24 run read the full-precision restart coordinates where MOSAICS and OpenMM read the three-decimal PDB, so it is a comparison of two geometries and is left out. The 1efs CHARMM36 cell is the 816-atom 5′-OH rebuild of the heteroduplex (F-08) and is not comparable with the AMBER rows above it. The two 1F7Y rows run on decks that exist only on an unmerged engine branch. Every run is a single-point energy at one fixed geometry, in vacuum, with no cutoff and dielectric 1.

Which chemistry the test systems reach. A deck is a table, and a single-point energy exercises only the rows the structure selects. The eleven systems reach all eight standard nucleic bases, DA, DC, DG and DT on the dodecamer and A, C, G and U on the tetraloop and the RNA 4-mer; 5′-OH and 3′-OH chain ends on five systems and 5′-phosphate interior chemistry on the padded heteroduplex; twenty amino-acid types across ubiquitin, BPTI and the MHC heavy chain, including charged, aromatic, histidine and proline residues, three disulfides, charged termini and two capped fragments. What no system reaches is a 5′-phosphate terminus under any deck, an alternative protonation state, or more than one conformation: each cell is one geometry, so a parameter that is wrong but inactive at that geometry, or a torsion whose phase is wrong but degenerate there, would not show. Finding F-21 is that failure mode caught in the act: an analytic force that was identically zero passed every energy comparison on the peptide.

3.2 Single-point agreement over 26 combinations

Table 3 gives every total energy as each program returns it, with the electrostatic term of MOSAICS and sander rescaled to the single constant of Section 3.4, and the difference of each reference from MOSAICS. Figure 2 puts all three programs on one reference so the MOSAICS spread can be read against the spread the two established engines show between themselves, and Figure 3 resolves the comparison into terms wherever the engine output files carry the split.

Nineteen of the 26 cells agree with every reference available to them to within 0.001 kcal/mol, and thirteen

Table 3. Single-point total energies on the eleven systems, in kcal/mol, vacuum, no cutoff, every electrostatic term on the CODATA-derived constant 332.0637133. A dash means that reference was not run for that cell, for the reason given in Section 3.1. The right-hand block is the reference minus MOSAICS. The 1efs CHARMM36 row is the 816-atom 5'-OH rebuild of the heteroduplex, not the 818-atom structure of the rows above it; the 1F7Y decks are on an unmerged engine branch. Source: `work/preprint-I/energies/MATRIX.json`, and the engine output files it names; the table is written from that file by `make_tables.py`.

System	Force field	Total energy (kcal/mol)				Reference minus MOSAICS (kcal/mol)		
		MOSAICS	sander	OpenMM	GROMACS	sander	OpenMM	GROMACS
1efs	OL21/OL3	2526.186378	2526.186713	2526.186575	2526.186580	+0.000335	+0.000197	+0.000202
	OL15/OL3	2522.856302	2522.856613	2522.856501	2522.856506	+0.000311	+0.000199	+0.000204
	OL24/OL3	2954.803590	—	2954.803568	—	—	-0.000022	—
	CHARMM36	3775.995925	—	3775.995927	—	—	+0.000002	—
1BNA	OL21	2141.488688	2141.487481	2141.487431	2141.488017	-0.001208	-0.001258	-0.000671
	OL15	2138.533139	2138.531881	2138.531881	2138.532468	-0.001258	-0.001258	-0.000671
	CHARMM36	3509.022307	—	3509.022309	—	—	+0.000001	—
	OL24	2954.122454	—	2954.122861	—	—	+0.000407	—
d(ACGT)	OL21	942.827346	942.827874	942.827764	—	+0.000528	+0.000418	—
	OL15	945.670885	945.671374	945.671303	—	+0.000489	+0.000418	—
	OL24	1080.088750	—	1080.088749	—	—	-0.000000	—
2KOC	OL3	347.450922	347.450964	347.451027	347.450987	+0.000042	+0.000106	+0.000065
	CHARMM36	2106.624466	—	2106.624466	—	—	+0.000000	—
r(ACGU)	OL3	2584.335888	2584.336025	2584.335946	—	+0.000137	+0.000058	—
1UBQ	ff14SB	-546.960403	-546.959674	-546.959594	-546.959625	+0.000729	+0.000809	+0.000778
	ff19SB	-1022.050252	-1022.050874	-1022.050802	—	-0.000622	-0.000550	—
5PTI	ff14SB	-332.312157	-332.311788	-332.311563	-332.311590	+0.000369	+0.000594	+0.000567
	ff19SB	-688.753926	-688.711488	-688.711342	—	+0.042438	+0.042584	—
ACE-AGSV-NME	ff14SB	96.262476	96.262430	96.262484	96.262481	-0.000046	+0.000008	+0.000005
	ff19SB	69.244797	69.244830	69.244805	—	+0.000033	+0.000008	—
Tax 11–19	ff14SB	-23.358508	-23.358422	-23.358416	—	+0.000087	+0.000092	—
	ff19SB	-82.278687	-82.278622	-82.278606	—	+0.000066	+0.000081	—
1HHK	ff14SB	-475.878441	-475.874656	-475.873684	—	+0.003785	+0.004757	—
	ff19SB	-2937.222999	-2937.219256	-2937.218706	—	+0.003743	+0.004293	—
1F7Y	ff14SB/OL3	14238.328640	14238.329782	14238.329881	—	+0.001142	+0.001241	—
	ff19SB/OL3	13685.351529	13685.353082	13685.353068	—	+0.001553	+0.001539	—

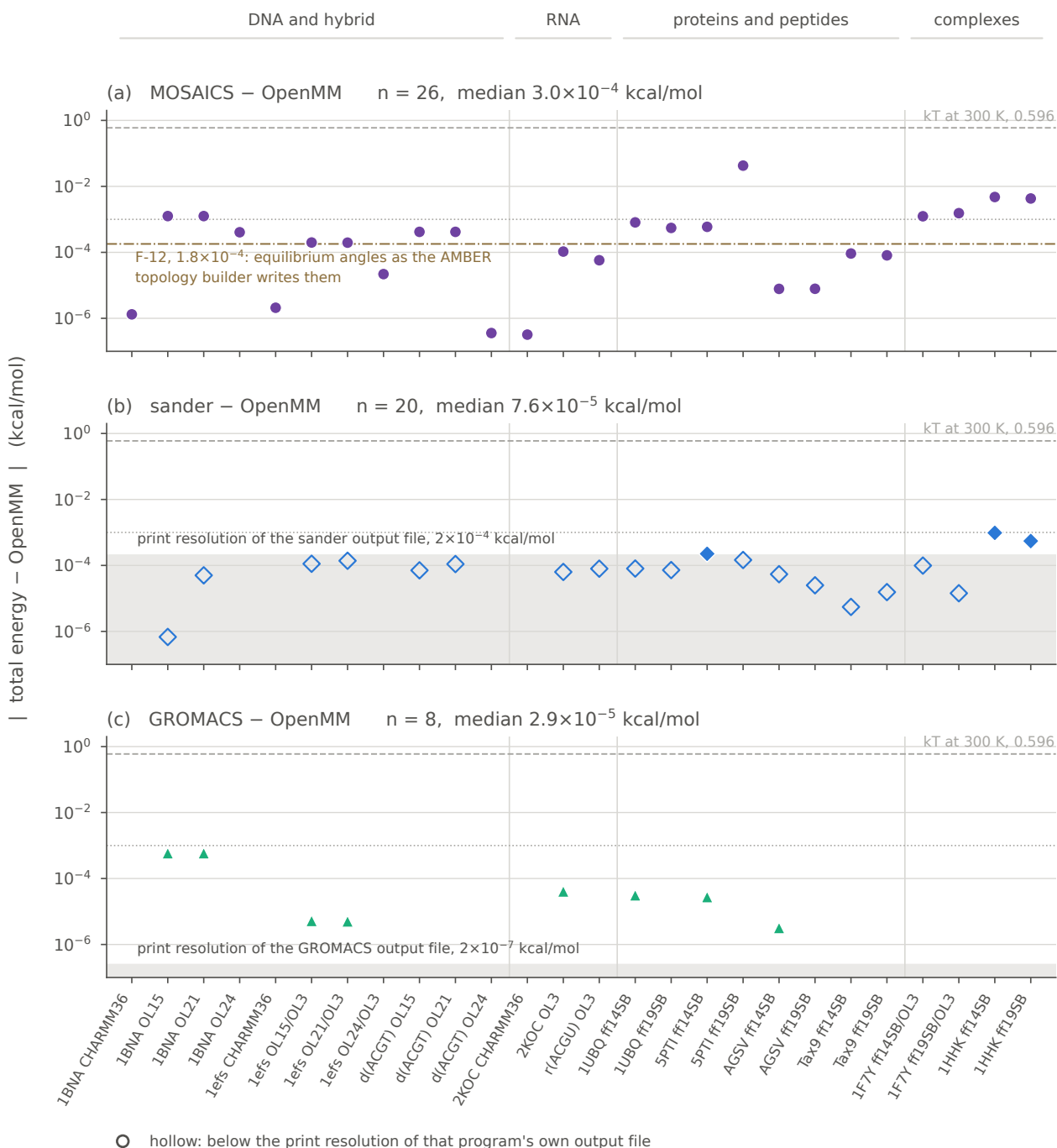


Figure 2. The three programs measured against a common reference, one panel each, over the 26 combinations grouped by system class. The vertical axis is $|E_{\text{program}} - E_{\text{OpenMM}}|$ on a log scale, the same scale in all three panels, so the panels may be read against one another. OpenMM is the reference for three reasons the panels themselves show. Its writer resolves to 5×10^{-13} kcal/mol, whereas sander prints four decimals and returns a total that is the sum of those rounded terms, so a sander difference carries up to 2×10^{-4} kcal/mol of print quantisation, shaded in (b); it ran on every cell, whereas sander was not run on the CHARMM36 cells, which are the closest agreement here; and it shares its Coulomb constant with GROMACS, so only sander needs the rescale of Section 3.4. Hollow markers lie below the print resolution of that program's own output file and are quantisation, not measured disagreement. Dashed line, kT at 300 K, 0.596 kcal/mol; dotted line, 0.001 kcal/mol. The dot-dashed line in (a) is finding F-12: the AMBER topology builder writes every equilibrium angle as the published value times a seven-digit approximation of $\pi/180$ while MOSAICS reads the exact value, which is worth 1.8×10^{-4} kcal/mol on 1efs and is the largest remaining difference on every AMBER row that passes. There the references carry the error, not MOSAICS, and the three CHARMM36 cells, where it does not arise, agree to 3×10^{-7} to 2×10^{-6} kcal/mol. Every value is re-derived from data/MATRIX.json by figures/energy-vs-openmm.py.

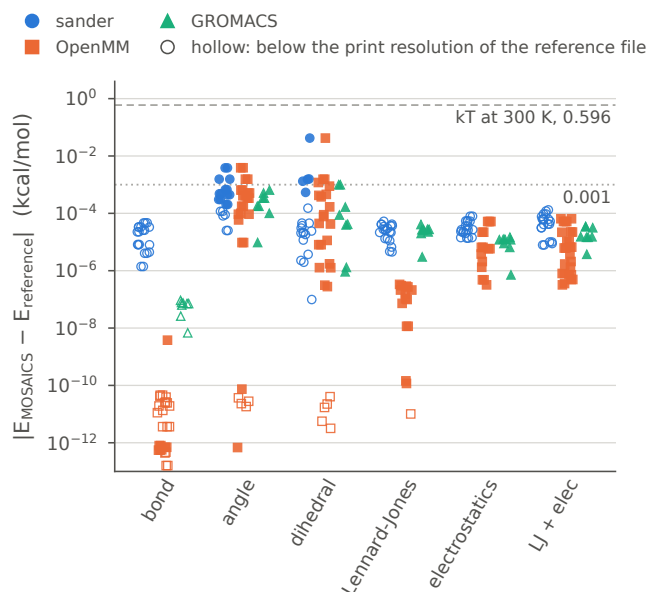


Figure 3. The same comparisons resolved into the terms the engines share, for the cells whose output files carry the split; where an OpenMM file prints only a combined non-bonded energy the comparison appears in the last column. Marker shape and colour identify the reference engine, and the vertical axis is $|E_{\text{MOSAICS}} - E_{\text{reference}}|$. Hollow markers fall below the print resolution of the reference file they were read from. The angle column carries the F-12 offset of Figure 2, which is why it sits above the others. Dashed line, kT at 300 K; dotted line, 0.001 kcal/mol. Every value is re-derived from `MATRIX.json` and the engine outputs it names by `figures/energy-agreement.py`.

to within 0.0004. Over the 26 cells the median of the largest gap to any reference is 0.00037 kcal/mol; the smallest is 3.2×10^{-7} , on the tetraloop under CHARMM36, and the largest is 0.0426, on BPTI under ff19SB. For scale, kT at 300 K is 0.596 kcal/mol, and the sander column can resolve nothing below about 2×10^{-4} . The seven cells outside 0.001 kcal/mol are, in order, BPTI ff19SB (0.0426), the two peptide-MHC cells (0.0048 and 0.0043), the two S15-rRNA cells (0.0012 and 0.0016) and the dodecamer under OL21 and OL15 (0.0013 each). Each has a mechanism, and they are of three kinds.

An improper read in two atom orders. The BPTI ff19SB cell is 0.0426 kcal/mol below the references on the dihedral term and nowhere else: bond, angle, Lennard-Jones and electrostatics carry the same residuals as the ff14SB cell on the same coordinates, and the correction map matches OpenMM to 5.8×10^{-13} . Splitting the dihedral term puts the whole gap in the impropers, 5.5085 against 5.5506 kcal/mol. It is one improper, at the backbone nitrogen of ASP 3. The AMBER topology writes that improper in the order C-CA-N-H and the MOSAICS residue template in the order -C-H-N-CA; the two orders define different dihedral angles, which coincide when the nitrogen is planar and part company when it is not, and at ASP 3 in the deposited coordinates they read -156.8° and $+158.4^\circ$, some 22° from planar. Evaluating that one improper in the topology’s order, with the two conventions for the phase and the deck’s coefficient rounding accounted for, reproduces the whole gap to 8.5×10^{-8} kcal/mol, and the accounting is committed beside the run. The same mechanism produced a 0.46 kcal/mol gap at LEU 2 of the Tax nonamer on an earlier build of that structure, which is why the peptide’s first amide hydrogen is now placed in the peptide plane by the engine’s builder and why the nonamer cells in Table 3 agree to 9×10^{-5} . The ff14SB cell on the same BPTI coordinates is within 6×10^{-4} , so the parameters are not in question: the improper order is.

Six adenine impropers on the dodecamer. Under OL21 and OL15 the dodecamer sits 0.0013 kcal/mol above sander and OpenMM and 0.0007 above GROMACS. Resolved into terms the excess is on the torsion term, $+1.5 \times 10^{-3}$, of which $+1.0 \times 10^{-3}$ is improper, and it was located rather than guessed. `tleap` under DNA.OL21 writes the adenine N9 improper as C8/C4/N9/C1’ and under ff99bsc0 as C4/C8/N9/C1’; swapping the first two atoms is not a symmetry of the term when the group is not exactly planar; MOSAICS agrees with the parmbsc0 order to eight figures, and the six instances that differ, on DA 5, 16 and 17, account for the

whole 1.0×10^{-3} . The same term on the heteroduplex, whose entire improper energy is 5.3×10^{-4} kcal/mol, is within 4.3×10^{-5} , so the difference between the two structures is the geometry the improper is evaluated at, not the parameters.

Two large systems not yet resolved into terms. The peptide-MHC complex, 6,139 atoms, is 0.0038 kcal/mol above sander and 0.0048 above OpenMM under ff14SB, and 0.0037 and 0.0043 under ff19SB, out of a total of -476 and -2937; per atom that is below 10^{-6} kcal/mol, and the two force fields give the same offset within 5×10^{-4} , which points at the non-bonded or angle terms they share rather than at the backbone terms they do not. The S15-rRNA complex, 3,321 atoms, is 0.0012 and 0.0016 kcal/mol below the references, and its terms are resolved: bond agrees to 2×10^{-11} , the correction map to 2×10^{-11} , torsion and improper to 4×10^{-4} and 8×10^{-5} , while the angle term carries -1.55×10^{-3} and the non-bonded sum -3.7×10^{-3} in the two directions that nearly cancel. Those two systems are the only ones whose reference coordinates were written from the three-decimal PDB rather than from tleap's restart, and the S15-rRNA decks are on a branch; both cells are reported as they stand and neither is claimed to be closed.

The angle term, on every system. The one residual that appears on every system whose terms were resolved is on the angle term, and it always has the same sign. MOSAICS minus OpenMM on that term is -1.76×10^{-4} kcal/mol on the heteroduplex, -9.6×10^{-6} on the tetrapeptide, -1.0×10^{-4} on the tetraloop, -3.4×10^{-4} on the dodecamer, -5.1×10^{-4} on BPTI and -6.5×10^{-4} on ubiquitin, growing with the number of angles. It is not a MOSAICS error. The AMBER topology builder writes every equilibrium angle as the published value multiplied by a seven-digit approximation of $\pi/180$, while MOSAICS reads the value in degrees and converts it exactly, so the reference carries the rounding (F-12). It is reported and not corrected, and it is the largest residual on every AMBER cell that is inside 0.001 kcal/mol.

3.3 Term by term on the two reference systems

Before the numbers, the resolution of the files they are read from, because several of the differences below are smaller than the output they come out of. sander prints four decimals, an absolute floor near 2×10^{-4} kcal/mol. GROMACS energies are read from sp.xvg, which carries six decimals in kJ/mol, a floor of 2.4×10^{-7} kcal/mol. OpenMM's single-point output carries twelve decimal places, a floor of about 5×10^{-13} kcal/mol. MOSAICS prints sixteen significant digits. A difference smaller than the floor of the file it is taken from is the last printed digit of that file and not a measurement, and no such number is quoted here as a residual.

On every one of the six original AMBER combinations, the largest difference between MOSAICS and OpenMM on any energy term is 1.8×10^{-4} kcal/mol for the four heteroduplex decks and 9.6×10^{-6} kcal/mol for the two peptide decks, in both cases on the angle term and for the reason just given. Every other term on every other row is smaller than this. The margin varies widely: the parmbse1 dihedral residual, $+1.03 \times 10^{-4}$ kcal/mol, is within a factor of two of it, while the two peptide bond terms agree past the twelfth decimal place OpenMM writes, which is as far as the comparison can be taken and no further. Table 4 gives the full four-engine breakdown for one force field, OL21/OL3 on 1efs, so that the raw numbers are on the page, and Table 5 gives MOSAICS minus each reference, per term, for all six combinations.

On the two non-bonded terms, the reference engines disagree with each other more than MOSAICS disagrees with either. OpenMM and GROMACS are independently written codes, and on 1efs they were given the same parameters and the same coordinates. They differ from one another by $+2.63 \times 10^{-5}$ kcal/mol on Lennard-Jones and -3.40×10^{-5} on electrostatics, on all four nucleic decks. MOSAICS minus OpenMM on the same two terms is $+2.89 \times 10^{-7}$ on Lennard-Jones under OL21/OL3 ($+3.81 \times 10^{-6}$ under the two bsc decks) and $+2.20 \times 10^{-5}$ on electrostatics, the latter with both sides corrected to OpenMM's own printed constant. That pattern, bonded terms with few parameters surviving and non-bonded terms with thousands not, is what a text-format topology conversion would produce, since a GROMACS .top stores fewer digits per parameter than the binary topology the other two engines read. We have not tested that attribution directly. What the numbers establish is a bound that has nothing to do with MOSAICS: two mature MD codes, handed the same force field through the ordinary conversion route, land 3×10^{-5} kcal/mol apart on an 818-atom

Table 4. OL21/OL3 on 1efs (818 atoms, net charge -26 , vacuum, no cutoff), in kcal/mol, as each engine prints them. The dihedral row includes impropers, and the Lennard-Jones and electrostatic rows include the 1–4 terms. sander gives four decimals (floor 2×10^{-4} kcal/mol), the GROMACS column is converted from sp.xvg values carrying six decimals in kJ/mol (floor 2.4×10^{-7}), OpenMM writes twelve decimals (floor 5×10^{-13}) and MO-SAICS writes sixteen significant digits. *corrected* means the electrostatic term rescaled to the single constant of Section 3.4; the MOSAICS column is divided by the unrounded product $0.529177 \times 627.50921$ and the sander column by $18.2223^2 = 332.05221729$, so either corrected entry can be reproduced digit for digit from the printed one above it. Sources: evidence/1efs/ol21_ol3/{mosaics,sander,openmm,gromacs}/.

Term	MOSAICS	sander	OpenMM	GROMACS
bond	70.99877741	70.9988	70.99877741	70.99877749
angle	220.56489609	220.5651	220.56507252	220.56507290
dihedral	596.17955421	596.1796	596.17959679	596.17959369
Lennard-Jones	-219.29643681	-219.2964	-219.29643710	-219.29646343
electrostatic, printed	1857.73806648	1857.6753	1857.73956494	1857.73959895
electrostatic, corrected	1857.73958864	1857.73961475	1857.73956494	1857.73959895
total, corrected	2526.18637954	2526.18671475	2526.18657455	2526.18657959

system, and on the two non-bonded terms MOSAICS sits below that bound against either reference.

The peptide test is a targeted test of what ff19SB adds. Between ff14SB and ff19SB on the same capped peptide, the bond, angle, Lennard-Jones and electrostatic terms are bit-identical in all four engines. MO-SAICS returns the same bond (0.14749181), bend (0.98732525), Lennard-Jones (134.50567845) and electrostatic (-71.35989834 kcal/mol) energies in both runs, agreeing to the last of the sixteen digits it prints. The whole ff14SB-to-ff19SB difference is -27.01767928 kcal/mol, of which -25.09643922 is the backbone torsion and -1.92124006 is the correction map; every other term contributes exactly zero. The ff19SB row is therefore the correction-map machinery applied on top of an already-verified ff14SB, tested with everything else held fixed, and four residues select four of the sixteen maps the deck loads. On ubiquitin, BPTI and the two complexes the same machinery is exercised on every amino-acid type the deck holds, and the correction-map term matches OpenMM to between 3×10^{-13} and 2×10^{-11} kcal/mol on all six ff19SB cells.

The nucleic decks are different parameter sets, and each is reproduced as it stands. All three references return the same bond, angle, Lennard-Jones and electrostatic energies for all four AMBER nucleic decks on 1efs; only the dihedral term moves. That is the design of those decks showing through, and it makes the dihedral row of Table 5 the row that carries the result: MOSAICS follows each of the four dihedral values to between 3.6×10^{-5} and 1.0×10^{-4} kcal/mol against OpenMM and GROMACS, without being retuned for any of them. On the tetraloop the OL15/OL3 and OL21/OL3 terminal decks return the same energy, 347.4514261182977 kcal/mol before the constant correction, to all sixteen printed digits on every term, because OL15 and OL21 touch DNA torsions only and a pure RNA chain never reaches them; Table 3 carries that one measurement once, as OL3. The size of the gap between two decks’ absolute torsion energies is not reported here and is not a quantity this protocol can produce.

CHARMM36 nucleic acid, one reference. An OpenMM reference for the CHARMM36 nucleic deck was built from charmm36_2024.xml and evaluated on the two free-ended 4-mers, the dodecamer, the tetraloop and the 5’-OH rebuild of the heteroduplex. The deck covers 24 residues (every base as a 5’, internal and 3’ unit, DNA and RNA) with 237 bonds, 562 bends, 1036 torsion entries and 15 753 non-bonded pairs. Table 6 gives the two 4-mers term by term. Every bonded term matches to the digits OpenMM prints, and that includes the two CHARMM terms the AMBER force fields do not have, the Urey–Bradley 1–3 harmonics and the harmonic impropers. What the improper row shows needs care: its total is 1.0×10^{-3} kcal/mol over 188 terms, so the structure sits at the improper minimum for practical purposes, and that is exactly where the engine’s truncated sixth-order expansion and CHARMM’s quadratic coincide by construction. Ten-digit agreement there confirms that the 188 entries were converted and selected correctly; it does not exercise the functional form away from equilibrium. The electrostatic term is the only one that does not match as printed,

Table 5. MOSAICS minus each reference engine, per term, in kcal/mol, on the six original combinations. Electrostatics are corrected on both sides to $332.0637133 \text{ kcal } \text{\AA} \text{ mol}^{-1} e^{-2}$ (Section 3.4) before the difference is taken, with the MOSAICS side divided by the unrounded $0.529177 \times 627.50921 = 332.06344122017$. *n.r.* means the difference is smaller than the resolution of that reference’s own output file, so the value in the committed evidence is a print artefact; the 27 censored sander entries run from 2.0×10^{-6} to 5.1×10^{-5} in the raw files. The GROMACS CMAP entry is the conversion-route limit F-06 and not a disagreement about physics. – marks a term the force field does not have. Source: evidence/results.txt and the per-combination output files under evidence/.

Term	OL21/OL3	OL15/OL3	parmbsc1	parmbsc0	ff14SB	ff19SB
<i>Reference: OpenMM 8.5.2, output floor $5 \times 10^{-13} \text{ kcal/mol}$</i>						
bond	n.r.	$+3.78 \times 10^{-9}$	-8.30×10^{-8}	-8.30×10^{-8}	n.r.	n.r.
angle	-1.76×10^{-4}	-1.76×10^{-4}	-1.76×10^{-4}	-1.76×10^{-4}	-9.57×10^{-6}	-9.57×10^{-6}
dihedral	-4.26×10^{-5}	-4.45×10^{-5}	$+1.03 \times 10^{-4}$	$+3.59 \times 10^{-5}$	$+1.27 \times 10^{-6}$	$+1.28 \times 10^{-6}$
cmmap	–	–	–	–	–	-5.69×10^{-8}
Lennard-Jones	$+2.89 \times 10^{-7}$	$+2.71 \times 10^{-7}$	$+3.81 \times 10^{-6}$	$+3.81 \times 10^{-6}$	$+1.15 \times 10^{-8}$	$+1.15 \times 10^{-8}$
electrostatic	$+2.37 \times 10^{-5}$	$+2.37 \times 10^{-5}$	$+2.37 \times 10^{-5}$	$+2.37 \times 10^{-5}$	$+4.18 \times 10^{-7}$	$+4.18 \times 10^{-7}$
<i>Reference: sander 24.0, output floor $2 \times 10^{-4} \text{ kcal/mol}$</i>						
bond	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
angle	-2.04×10^{-4}	-2.04×10^{-4}	-2.04×10^{-4}	-2.04×10^{-4}	n.r.	n.r.
dihedral	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
cmmap	–	–	–	–	–	n.r.
Lennard-Jones	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
electrostatic	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
<i>Reference: GROMACS 2025.4, output floor $2.4 \times 10^{-7} \text{ kcal/mol}$</i>						
bond	n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
angle	-1.77×10^{-4}	-1.77×10^{-4}	-1.77×10^{-4}	-1.77×10^{-4}	-9.60×10^{-6}	-9.60×10^{-6}
dihedral	-3.95×10^{-5}	-4.16×10^{-5}	$+1.07 \times 10^{-4}$	$+3.90 \times 10^{-5}$	$+1.30 \times 10^{-6}$	$+1.38 \times 10^{-6}$
cmmap	–	–	–	–	–	-3.02×10^{-1} (F-06)
Lennard-Jones	$+2.66 \times 10^{-5}$	$+2.66 \times 10^{-5}$	$+3.01 \times 10^{-5}$	$+3.01 \times 10^{-5}$	$+3.02 \times 10^{-6}$	$+3.02 \times 10^{-6}$
electrostatic	-1.03×10^{-5}	-1.03×10^{-5}	-1.03×10^{-5}	-1.03×10^{-5}	$+6.49 \times 10^{-7}$	$+6.49 \times 10^{-7}$

at $9.8 \times 10^{-5} \text{ kcal/mol}$ on the RNA 4-mer and 2.9×10^{-4} on the DNA 4-mer, and the charge-to-energy constant accounts for that offset to four significant figures: corrected exactly as the AMBER rows are corrected, the two residuals fall to -2.4×10^{-10} and $-9.0 \times 10^{-10} \text{ kcal/mol}$. On the three larger structures the corrected totals differ from OpenMM by -3.5×10^{-7} (dodecamer), $+3.2 \times 10^{-7}$ (tetraloop) and $-2.1 \times 10^{-6} \text{ kcal/mol}$ (816-atom heteroduplex). That is the closest agreement anywhere in this work, and Section 4.5 says why.

3.4 The charge-to-energy constant, and why every electrostatic number is corrected

The four programs do not use the same constant to turn charges into an energy. This was measured and not assumed, by the two-charge topology of Section 2.6, and Table 7 gives the four values. Every electrostatic term in this work is put on one target constant before any difference is taken, and that target has to be a number with a derivation, or the correction becomes a preference. It is

$$\frac{e^2 N_A}{4\pi\epsilon_0 \times 1 \text{ \AA} \times 4.184 \text{ J cal}^{-1}} = 332.0637133 \text{ kcal } \text{\AA} \text{ mol}^{-1} e^{-2},$$

evaluated with the CODATA 2018 recommended values[53]. The elementary charge, $e = 1.602176634 \times 10^{-19} \text{ C}$, and the Avogadro constant, $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$, are both exact by the 2019 SI definitions; $\epsilon_0 = 8.8541878128 \times 10^{-12} \text{ F m}^{-1}$; and the thermochemical calorie is exactly 4.184 J. The 2022 revision of ϵ_0 gives 332.0637131 instead, lower by 6.8×10^{-10} relative, which is worth $1.3 \times 10^{-6} \text{ kcal/mol}$ on the 1e fs electrostatic term.

sander is the outlier, not MOSAICS. It stores topology charges pre-multiplied by 18.2223 and its constant is that number squared, which is an older value; OpenMM and GROMACS, written independently of each other, agree with each other and with the derived target to every decimal their measurement was recorded

Table 6. CHARMM36 nucleic acid against OpenMM 8.5.2, in kcal/mol, on the two free-ended 4-mers with 5TER/3TER termini in vacuum, as printed. r(ACGU) is 127 atoms and d(ACGT) is 126. Urey-Bradley is CHARMM’s 1–3 harmonic term, which in OpenMM sits inside HarmonicBondForce and is separated here. The two d(ACGT) bonded columns are inflated by a defect in that committed structure, in which one H–C–H angle is 3.15° instead of 109.5° (F-17, Section 4.6); the comparison between the two programs is unaffected, because both read the same coordinates. The electrostatic rows differ by the charge-to-energy constant and by little else. Sources: evidence/charmm36_rna_acgu.tsv, evidence/charmm36_dna_acgt.tsv.

Term	r(ACGU)		d(ACGT)	
	MOSAICS	OpenMM	MOSAICS	OpenMM
bond	28.51501806	28.51501806	18.92184909	18.92184909
Urey-Bradley	27.33939420	27.33939420	167.65137978	167.65137978
angle	166.36120948	166.36120948	412.78809431	412.78809431
torsion	114.02952401	114.02952401	122.94661546	122.94661546
improper	0.00102252	0.00102252	0.00102555	0.00102555
1–4 Lennard-Jones	230.89575106	230.89575106	1412.27140098	1412.27140098
Lennard-Jones	2564.74970076	2564.74970076	176.68820406	176.68820406
electrostatic	–119.98431509	–119.98441340	–356.57731681	–356.57760898
total	3011.90730500	3011.90720669	1954.69125242	1954.69096025

Table 7. The charge-to-energy constant of each program, in kcal Å mol^{–1} e^{–2}. sander, OpenMM and GROMACS were measured with two unit charges 1 Å apart in vacuum, everything else switched off; the MOSAICS entry is evaluated from the constants compiled into the engine, $0.529177 \times 627.50921$. sander’s measured value is limited by its four-decimal print precision; its exact value is $18.2223^2 = 332.05221729$, and that full value is what corrects the sander column. The OpenMM and GROMACS measurements were recorded to six decimals, which bounds their relative difference from the target at 1.5×10^{-9} and says nothing finer. Source: evidence/coulomb-constants/.

Engine	Constant	Relative to sander	Relative to 332.0637133
sander	332.052200	0	-3.5×10^{-5}
OpenMM	332.063713	$+3.467316 \times 10^{-5}$	$< 1.5 \times 10^{-9}$, print-limited
GROMACS	332.063713	$+3.467284 \times 10^{-5}$	$< 1.5 \times 10^{-9}$, print-limited
MOSAICS	332.063441	$+3.385377 \times 10^{-5}$	-8.2×10^{-7}

to. MOSAICS sits 8.2×10^{-7} below the target because three constants compiled into the engine are rounded (F-13). The size of this effect exceeds every difference the comparison is trying to resolve. On 1efs the sander electrostatic term moves by +0.0643147 kcal/mol when rescaled, and the MOSAICS term by $+1.522158 \times 10^{-3}$; the sander shift is roughly 360 times the largest residual left after correction and about a tenth of kT at 300 K. An uncorrected cross-engine comparison of an 818-atom charged system would therefore report a disagreement between sander and OpenMM bigger than everything else on the page, and it would be reporting a unit convention.

Which constant is the target has to be stated, because the choice is worth something. Corrected to OpenMM’s own printed constant instead of the derived one, MOSAICS minus OpenMM on electrostatics is $+2.202 \times 10^{-5}$ kcal/mol on all four 1efs rows against the $+2.370 \times 10^{-5}$ of Table 5; the 1.68×10^{-6} between the two frames is what a six-decimal printed constant costs, and neither choice changes any conclusion. After correction, $+2.37 \times 10^{-5}$ kcal/mol remains on the 1efs electrostatic term, 1.3×10^{-8} of the term, and we cannot account for it. One hypothesis fits what is on the page. The AMBER decks reach MOSAICS through a text route, and a prmtop writes charges in %FORMAT(5E16.8), about 10^{-8} relative per charge, the same order as the residual; the CHARMM36 deck does not take that route, its charges are read out of a built OpenMM System at full double precision, and its corrected residuals are three orders of magnitude smaller on the same engine with the same correction. Testing this needs a diff of the deck charges against the prmtop charges at full precision, which is one command and is not in the evidence.

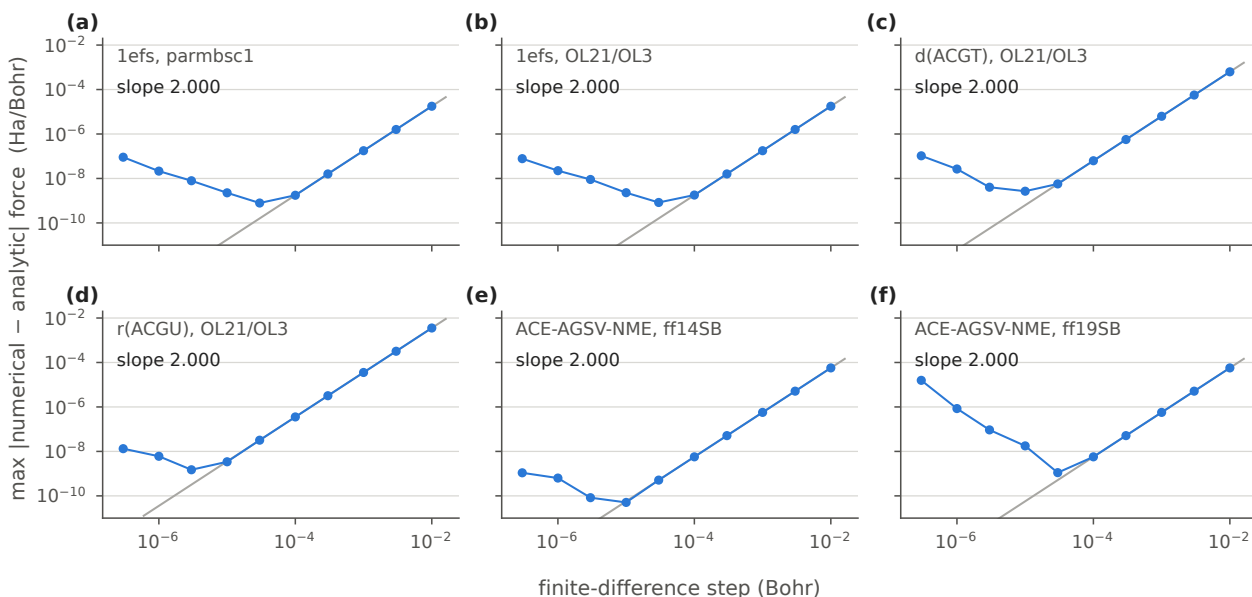


Figure 4. Convergence of the central-difference check on the total-energy gradient for six of the 61 sweeps, one per Cartesian system and force field. Each point is the largest $|\text{numerical} - \text{analytic}|$ force component over all atoms at that step. The grey line has slope exactly 2 through the coarsest point; the printed exponent is the least-squares fit over the fixed window 10^{-2} to 3×10^{-4} Bohr. Below the window each sweep leaves the falling branch where round-off in the energy overtakes the discretisation error, which is a floor and not a disagreement. Source: `evidence/forces/*.all.tsv`.

3.5 Forces

Analytic forces were checked against a central difference of the same energy, term by term, over a ten-point step sweep from 10^{-2} to 3×10^{-7} Bohr, on six Cartesian systems and, against a staging coordinate, on four staged systems: 61 sweeps in all, covering five terms and their sum on every system, plus the correction map on the ff19SB peptide. The residual is the largest $|\text{numerical} - \text{analytic}|$ over every atom and every component at a given step. Figure 4 plots the total-energy sweep on six of the systems.

A residual at a given step is not an error in the force. It is dominated by the discretisation error of the difference at coarse steps and by round-off in the energy evaluation at fine ones, and it is smallest where the two cross. At the coarsest step, 10^{-2} Bohr, residuals across the set run up to 3.6×10^{-3} Ha/Bohr, which is what a central difference of that width is expected to give. The number worth quoting is the minimum over each sweep. The largest of the 61 minima is 3.8×10^{-9} Ha/Bohr, on the non-bonded staging gradient of the RNA 4-mer; the smallest is 4.5×10^{-13} , on the peptide torsion under ff19SB. For scale, 0.002 kcal/mol per Bohr is 3.2×10^{-6} Ha/Bohr, and every one of the 61 minima is smaller than that by a factor of at least 800.

The exponent is the measurement, and it was fitted over a step window fixed in advance instead of chosen per sweep: 10^{-2} to 3×10^{-4} Bohr, the four coarsest steps, which every one of the 61 sweeps spans. Over that window the fitted exponents run from 1.994 to 2.009, with a mean of 2.000 and a standard deviation of 0.002. The fits are tight: the largest root-mean-square residual of any of the 61 is 0.003 in $\log_{10}$ of the force residual, and no single point departs from its own fit by more than 0.005. That is the second-order convergence a central difference gives when the analytic force really is the derivative of the energy. Below the window every sweep leaves the falling branch at its own step and flattens, which is round-off and not disagreement; the correction map leaves it earliest and climbs again, reaching 1.6×10^{-5} Ha/Bohr at 3×10^{-7} Bohr, because the map is interpolated on a grid and is not smooth to machine precision.

The sweep found one defect that no energy comparison in this paper could have found. The correction-map analytic force was identically zero at a dihedral of exactly π , which is where a peptide built extended sits; it is fixed and pinned by a test (F-21). The energy was never affected, and every sweep reported here is post-repair.

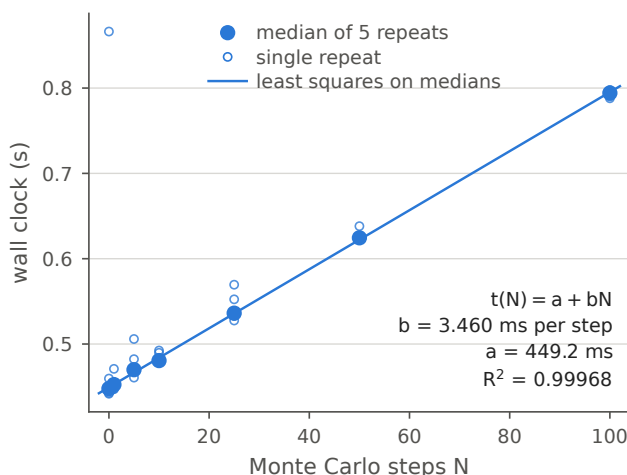


Figure 5. Wall clock against number of Monte Carlo steps for the committed 1efs OL21/OL3 input, 818 atoms, serial, one core of an Apple M5. Small hollow markers are the five repeats at each N , filled markers their medians, the line the least-squares fit to the medians. The one 866 ms outlier at $N = 0$ is why medians are used. Source: `work/preprint-I/runs/2026-08-31-cost-818/result.json`, re-fitted by `figures/cost-fit.py`, which exits if its fit disagrees with the committed one.

3.6 What one Monte Carlo step costs

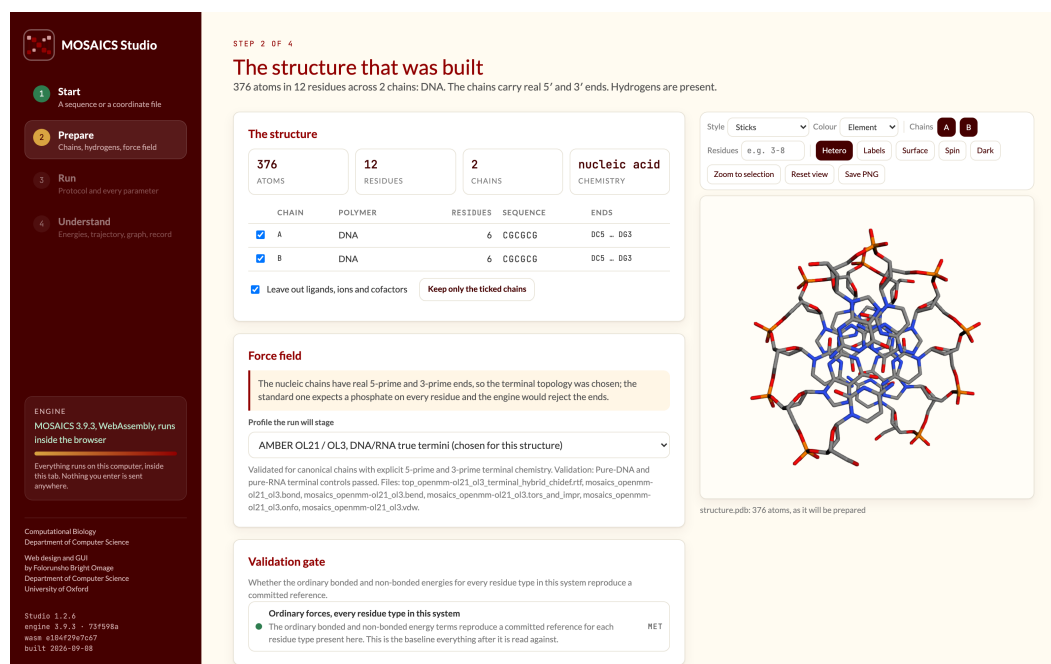
One Monte Carlo step on the 818-atom heteroduplex under OL21/OL3 costs 3.46 ms on one core of an Apple M5 laptop, with 449 ms of setup that happens once per run; the linear fit over $N = 0$ to 100 steps has $R^2 = 0.99968$ (Figure 5). A step is one torsional proposal plus one full energy evaluation over every non-excluded pair, so the figure is an upper bound on the energy evaluation alone; the proposal is inside it and is not separated here. It is a cost, not a benchmark: nothing here compares MOSAICS with another program on the same hardware, the engine has never been tuned for the speed of a single evaluation, and the number says nothing about scaling or about other sizes. What it settles is the question a reader of the 26-cell matrix is entitled to ask next, which is what one of those evaluations costs when the engine is actually sampling. The strength the method claims lies elsewhere, in the number of evaluations a guided proposal needs to cross a barrier that molecular dynamics must climb in femtosecond steps, and that ratio is not measured in this paper.

3.7 MOSAICS Studio: the same engine, verified in the browser

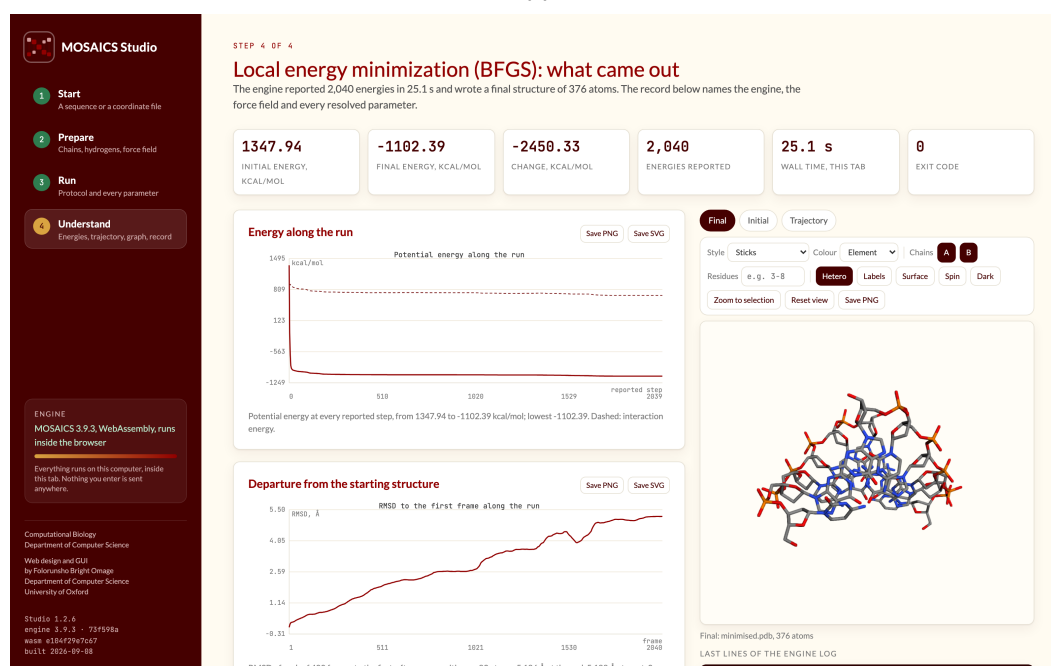
The Studio is only as good as the engine it carries, so its verification is of the same kind as the rest of this paper: the same input, run under the native engine and under the WebAssembly one, compared number by number. Figure 6 shows the page at two of its four steps on the run used for that check.

Native against WebAssembly. The Studio’s documented case study builds CGCGCG as a B-DNA duplex, minimises it with the `local-minimum` preset (BFGS to a gradient tolerance of 10^{-7}) on the OL21/OL3 terminal deck, exports a residue graph and writes the report. Replaying the deck the Studio writes under the native Apple-silicon engine and under the WebAssembly engine gives Table 8: the same exit code, the same 2,040 BFGS iterations, the same first and last energies to the two decimals the engine’s energy file carries, a largest difference along the series of 0.25 kcal/mol, and a final-structure RMSD of 0.0045 Å between the two. A headless test drives the whole pipeline, the same Python archive and the same bridge the page uses, and fails unless those numbers reproduce. The run in Figure 6 is that case study executed on the live page in a headless Chromium session on 8 September 2026: 2,040 energies reported, 1347.94 to −1102.39 kcal/mol, 25.1 s of wall time in the tab against 20.8 s under Node and 11.5 s for the native engine.

The record. The run in Figure 6(b) wrote a record naming Studio 1.2.6, the engine commit 73f598a and the SHA-256 of the WebAssembly build, the six staged files of the OL21/OL3 terminal deck by digest, the preset and every parameter it resolved, and the start and end times, with the final structure, the engine deck, the engine log, the energy series and the trajectory beside it, each one click away. That record is what a reader



(a)



(b)

Figure 6. MOSAICS Studio in the browser, version 1.2.6, on the CGCGCG duplex used for the cross-check of Table 8. (a) The Prepare step: the built structure, 376 atoms in 12 residues across two chains with real 5' and 3' ends; the force field chosen with its reason, here the OL21/OL3 terminal deck because the chains carry real ends; the validation gate reporting that the ordinary bonded and non-bonded energies for every residue type present reproduce a committed reference; and the 3Dmol.js viewer. The left rail names the engine as WebAssembly running inside the tab, with the Studio version, engine commit, engine digest and build date. (b) The Understand step after the BFGS minimisation: initial and final energies, the number of energies reported, the wall time in this tab and the exit code; energy along the run; RMSD to the first frame along the trajectory; and the final structure. The energy-against-RMSD plot, the residue-by-residue comparison, the graph export and the record follow further down the page. The Start and Run steps are in Supporting Information Figure S1.

Table 8. The CGCGCG case study replayed under the native engine and under the WebAssembly engine, from the Studio’s build record. The native binary is the Apple-silicon 3.9.1 build at engine commit ea00b02; the WebAssembly build is from commit 73f598a (7 September 2026). Source: `web/engine/BUILD.md` in the Studio repository.

	native, arm64	WebAssembly
exit code	0	0
BFGS iterations reported	2040	2040
first energy (kcal/mol)	1347.94	1347.94
last energy (kcal/mol)	-1102.39	-1102.39
largest difference along the series (kcal/mol)	0.25	
final-structure RMSD (Å)	0.0045	
wall time, Node 26 (s)	11.5	20.8

who repeats the run compares against, and it is the same document on every machine that loads the page, because no part of the computation happens anywhere else.

4 Discussion

4.1 What an additive all-atom energy makes available

The sampling half of natural-move Monte Carlo is unchanged by any of this. What changes is what an accepted structure is worth. A knowledge-based potential returns a ranking: it says which of two models is preferred, on a scale fitted to that comparison and to the representation it was fitted on [21, 22]. For structure prediction, and for refinement where an experimental restraint carries most of the signal [11, 12], that is sufficient. It is not sufficient for anything that needs the energy itself. Two such numbers computed for two different molecules do not lie on the same axis, nothing computed from the energy on an absolute scale is available, and a structure cannot pass between the Monte Carlo engine and a molecular dynamics run without the sampled quantity changing definition on the way.

With eight parameter families evaluated to the tolerances of Sections 3.2 and 3.3, those obstructions are lifted for the chemistry and the geometries tested, and for nothing beyond them. The claim the data supports is narrow and should be stated narrowly: handed a published deck, the engine reproduces that deck’s energy and its gradient to the tolerance reported, whichever deck it is. The four AMBER nucleic decks are substantially different parameter sets, and the engine follows each of them without being retuned for any. What that does not license is a statement about how far apart the decks put a molecule. A Fourier torsion sum, $\sum (V_n/2)[1 + \cos(n\phi - \gamma)]$, does not vanish at any reference state, and a refit moves its additive constant as freely as it moves its shape; the quantity that would carry that argument is $\Delta\Delta E$ between at least two conformations of the same molecule under two decks, and it is not computed here.

What the engine can now be asked, for the systems tested, is a different class of question. It can be run at a stated temperature against a named force field, so that an ensemble it produces and an ensemble a molecular dynamics code produces under the same parameters are ensembles of the same function; and a configuration produced by a collective move can be continued under molecular dynamics, or taken from one, without the energy being redefined in between. Neither is demonstrated in this paper. What is demonstrated is the precondition: the energy and its gradient are the published ones, at the geometries and on the chemistry tested.

4.2 What the extension to eleven systems adds, and what it does not

The two reference systems of Section 3.3 settle whether the engine evaluates a deck correctly, term by term, against three programs. The other nine settle a different question: whether it does so on the chemistry a user will actually bring. A deck is a few hundred to a few thousand parameter rows and a 56-atom peptide reaches a fraction of them. Ubiquitin, BPTI and the MHC heavy chain between them select every amino-acid type ff14SB and ff19SB carry, with charged, aromatic, histidine and proline residues, disulfides and charged

termini; the dodecamer and the tetraloop select all eight bases and both 5'-OH and 3'-OH ends; the two complexes select a bound peptide in a groove and a protein-RNA interface. That every one of those cells lands within a few thousandths of a kcal/mol of the references, and that the two that land further away resolve to the atom order of a single improper, is what the extension adds.

It also sharpens what the residuals mean. On the two reference systems the angle term carried the largest residual and it was attributed to the reference topology (F-12). Seven further systems now carry the same residual with the same sign, growing with the number of angles, which is what a rounding of $\pi/180$ in the topology builder predicts and what a defect in the engine's bend routine would not. The improper cells make the complementary point: the one large disagreement in the matrix is not a parameter or an expression but a convention, the order in which four atoms are listed, and it appears where the geometry departs from planarity and nowhere else. Cross-engine agreement certifies the evaluation path. It does not certify the path that produced the parameters, and both of the residuals that survive at the top of the matrix live there.

What the extension does not add is a per-term comparison against all three references for every cell. The totals are against two or three programs on 20 of the 26 cells and against OpenMM alone on the other six; the per-term resolution in Figure 3 exists only where the output files carry it, and eleven OpenMM comparisons print a single non-bonded sum. Nor does it add a second conformation per system. The extension widens the chemistry and holds the geometry fixed, and the failure mode F-21 demonstrates remains open on every cell.

4.3 What changed in the engine, and what did not

A reader is entitled to know how much of this is new code, because that is where the risk of a new defect lives. The honest answer is that most of it is not. Both CHARMM-specific terms were carried on machinery the engine already had. The 1497 Urey-Bradley entries in CHARMM36 reduce to 810 unordered triples, of which 807 are written into the bond slot of the existing bend record, which `bend.cpp` already evaluates as $\frac{1}{2}k(r_{13} - r_0)^2$, the functional form CHARMM asks for; three triples are dropped and counted, because four P-ON2-P variants carry negative force constants that CHARMM permits and the deck reader rejects, and they are pyrophosphate linkages that do not occur in a phosphodiester backbone (F-16). The 188 harmonic impropers are carried on the existing torsion record with `\pot_type{harm}`, which the deck reader expands onto the same power series it uses for the cosine form. That expansion is a genuine approximation, exact to four figures at 10° from equilibrium and -7.7% at 60° , and the ten-digit agreement reported for the improper term is at a total of 1.0×10^{-3} kcal/mol, where the series and the quadratic coincide by construction.

Where the defects were was the converter and, in a few places, the engine. The CHARMM XML splits the non-bonded term between two elements, charge in `NonbondedForce` and σ , ϵ in `LennardJonesForce`; the converter read only the first, so an entire CHARMM deck carried $\epsilon = 0$ on all 85 078 pair entries. The correction-map analytic force was identically zero at $\phi = \pi$, because a $\sin \phi$ clamp replaced a vanishing denominator and left a vanishing numerator (F-21); it is now written in the form with no $\sin \phi$ in it. Forty arrays allocated with `new[]` were grown with `realloc` (F-19), which ran on every committed job and was invisible because the relevant sanitizer check is off by default on macOS. And the deck reader now carries the per-molecule torsion label and chains the residues it builds; the released 3.9.1 refuses the decks that need the first and returns energies tens of thousands of kcal/mol from the references without the second.

The novelty is not all-atom Monte Carlo on an additive force field. That exists: CAMPARI with ABSINTH, MCPRO and BOSS, ProtoMS and Cassandra all evaluate all-atom additive potentials under Metropolis Monte Carlo, and concerted-rotation move sets have been run in OPLS for both protein and nucleic acid backbones [19, 20, 15]. What is new here is the pairing of that energy with the nested natural-move hierarchy and its stochastic chain closure [8, 9], so that one proposal can act at the scale of a domain and the next at the scale of a sugar pucker while the accepted energy stays a published one, and the delivery of that pairing to a user who has never seen the engine's input syntax. The claim is the combination, and it should be read that way.

4.4 Three references, and what the third one pays for

Agreement with a single reference program is a weaker claim than it looks. It establishes that two codes treat a parameter file the same way, which confounds correct evaluation with shared convention: the same choice of 1–4 scaling, the same grouping of impropers, the same charge-to-energy constant. Three independently written programs break that confound in both directions. Where they agree with one another, the target is not in dispute and a residual against any of them is a measurement of the implementation. Where they disagree, the disagreement is itself a number, and it bounds what any cross-engine claim can resolve. On 1efs the bound is 2.6×10^{-5} kcal/mol on Lennard-Jones and 3.4×10^{-5} on electrostatics, which is what OpenMM and GROMACS return against each other when handed the same parameters and the same coordinates through the ordinary conversion route.

Each of the two convention differences reported here was invisible to some subset of the three. The GROMACS route is the only one that could not carry ff19SB’s per-residue correction maps, and running it put a number on what that costs: 0.30 kcal/mol on a four-residue peptide and 8.4 kcal/mol on ubiquitin, from applying one residue’s map to every residue. sander, separately, is the only one of the three that does not share the modern charge-to-energy constant, and a comparison against OpenMM and GROMACS alone would have shown MOSAICS’s own 8.2×10^{-7} offset while leaving the 3.5×10^{-5} carried by a widely used engine undetected. Neither is a defect in the engine that exposed it, and neither would have been found by adding a fourth code from the same family. The independence is in the code and not in the input, and that limit is worth stating because it is where the argument is weakest: all three references read a topology written by one builder, so an error in the builder appears identically in all three, and F-12 is exactly that case. The three references are also not of equal weight: six combinations against three engines, fourteen more against two, and six against OpenMM alone are not the same strength of claim, and Table 3 keeps them apart.

4.5 The charge-to-energy constant is a general hazard

The constant that converts $q_i q_j / r$ into kcal/mol is not the same number in every program, and any group differencing energies across programs is exposed to that. sander uses $18.2223^2 = 332.052217$, an older value carried in the way it stores topology charges; OpenMM and GROMACS both return 332.063713 from the same two-charge measurement. The relative gap is 3.5×10^{-5} , which on the 1efs electrostatic term of 1857.7 kcal/mol is 0.064 kcal/mol, roughly 360 times the largest residual this work is trying to resolve and about a tenth of kT at 300 K. The constant enters multiplicatively, so it does not cancel when two conformations of the same molecule are compared: it biases the electrostatic part of that difference by the same 3.5×10^{-5} , and its absolute size grows with the electrostatic energy, which is to say with system size and with net charge. It also has nothing to do with the AMBER family: the same offset accounts for the CHARMM36 residuals on every one of the five CHARMM36 cells, and once it is removed those cells are the closest agreement in this work, at a few parts in 10^{12} of the term. The practical recommendation is to measure the constant instead of taking it from documentation, which is a few minutes of work in any engine that will read a topology, and to state which constant a comparison was corrected to, because the choice is worth 1.7×10^{-6} kcal/mol on this system.

4.6 What these numbers do not establish

The first limit is the one conformation. Every run sets all proposal widths to zero, so a term that happens to vanish at that geometry is untested. The capped peptide sits at the minimum of its out-of-plane terms and both MOSAICS and GROMACS return zero for them (F-11). More seriously, the correction-map analytic force was identically zero, 0 of 168 components, because a peptide built extended sits at $\phi = \psi = \pi$, the singular point of the implemented expression, and *every energy comparison on that system passed while that was true* (F-21). It took a gradient sweep to find it. The same failure mode is available to any term whose parameter is wrong but inactive, or whose phase is wrong but degenerate, at the one geometry tested, on any of the 26 cells. Repeating these single points over 50 to 100 decorrelated snapshots per system, as Shirts *et al.*[42] do, would close it, and it is cheap on vacuum systems of this size. It has not been done here.

The second is that the force sweeps cover four systems and the term-by-term comparison two. The nine

added systems carry totals, and per-term differences where the output files resolve them, but no force sweep and no staged sweep. The correction map is exercised at the force level on one system, because the ff19SB peptide is the only committed force input that loads a map database and no nucleic parameter set here carries one (F-22).

One committed structure is geometrically defective and the defect is not repaired. In `examples/pure_terminal_controls/dna_acgt.pdb`, residue DG 3, H5' and H5'' each sit 1.089 and 1.090 Å from C5', which is right, and 0.060 Å from each other, which makes the H–C–H angle 3.15° instead of 109.5° (F-17). The offending pair is 1–3, which the non-bonded sum excludes, so no energy check sees it as an error; the symptom is that this 126-atom 4-mer returns a bend energy of 508.62 kcal/mol under the AMBER decks against 220.56 for the 818-atom heteroduplex, and 412.79 under CHARMM36 against 166.36 for the RNA 4-mer of the same size. The cross-engine comparison is unaffected, because every program reads the same coordinates and they reproduce each other on them. The file was pinned in a test and left alone because committed reference tables were computed from it; repairing it and regenerating those tables is the right fix and has not been done.

Three scope limits belong with these. CHARMM36 for protein is not available in MOSAICS, and the obstruction is a file format and not a physical one: its correction maps are selected partly by the residue that follows the one being corrected, while the engine's map-selection record keys on the residue itself (F-09), and no claim about the CHARMM36 protein parameters [64, 65] appears anywhere in this work. Two decks cannot be regenerated from their sources (F-20), so the seventh decimal of the OL15/OL3 and OL21/OL3 bonded terms is reporting transcription; and the OL24 parameter files come from a package whose redistribution terms are its authors', so a reader who does not hold that package cannot rebuild those three cells from a clone. And `parmbsc1` has no terminal deck and `parmbsc0` does not run on a free-ended duplex, which is why the dodecamer has no cell under either.

There is a gap between the code path validated here and the code path a production run executes, and it is narrower than that sentence usually implies. MOSAICS scores a proposal by recomputing the whole potential, so the term routines checked here are the term routines a proposal is judged by. What no single point exercises is a choice of pair set: `inter_list` is none in every committed input, so the link-list and neighbour-list branches never ran. Closing that costs one single point per branch with a list cutoff longer than the molecule, differenced against the numbers reported here to the summation-order tolerance of 5.5×10^{-11} kcal/mol. Neither run is in this paper.

Single-point energies and finite-difference force checks fix the value of the energy function at a geometry and its gradient there. They say nothing about whether the move set, the stochastic chain closure and the acceptance rule together generate the Boltzmann distribution of that energy [8, 61]; that is a separate validation with separate diagnostics, and it is not attempted here. The cost figure of Section 3.6 is likewise not an efficiency result: a per-step time on one machine compares nothing with anything, and the quantity that would, sampling per unit cost against a molecular dynamics baseline on the same system and the same force field, is the next measurement and not this one. Nor do these numbers speak to whether ff19SB or OL21/OL3 is right about a molecule, which is measured against experiment and not against another program [62, 63, 35]. And everything was run in vacuum at dielectric 1 with no cutoff, so no cutoff scheme, switching function or implicit-solvent term is covered by any number in this paper.

4.7 What the Studio does and does not establish

The Studio's verification is of the engine it carries, on one documented run. The cross-check shows that the WebAssembly build follows the native one through 2,040 minimisation steps to the same energies at the precision the energy file carries and to 0.0045 Å in the final structure; it does not show that every protocol behaves identically in the browser, and two protocols are known to differ by construction, because the browser engine is single-threaded and parallel-tempering replicas take turns. The record binds what ran to what was produced; it does not make the run correct, and the validation gate on the Prepare step checks the ordinary energy terms of the residue types present against a committed reference, which is the baseline of this paper and not more. No external user has run the software at the time of writing, and proteins are prepared from coordinates rather than built.

4.8 Next

The intended application is conformational sampling of peptide–MHC complexes, and it is the next piece of work rather than a result reported here. The system suits the move set: a bound peptide with a small number of internal degrees of freedom in a groove, flanked by domains that move as units, which is the granularity natural moves were built for [9], and hierarchical natural-move Monte Carlo has already been applied to peptide dissociation from MHC class I [14]. What exists in this work is the implementation side of that study: the complex evaluates under both protein force fields to within 0.005 kcal/mol of two references at 6,139 atoms, the staging transform was checked on a peptide–MHC fixture at the gradient level, and the Studio will prepare and run it. No sampling run of any kind on that system appears in the evidence.

Several things should be settled before such a study is worth reporting, and each is small and named. The single points should be repeated over an ensemble of decorrelated configurations per system, the only cheap defence against the failure mode F-21 demonstrates. The force sweeps should be extended to the nine added systems. The two large cells that are not resolved into terms should be. The link-list and neighbour-list branches of the non-bonded sum need the two single points described above. The defective control structure should be repaired and its reference tables regenerated (F-17); the two decks that cannot be regenerated should be rebuilt (F-20); the peptide needs a deliberately distorted copy so that the out-of-plane terms and the CHARMM improper series are exercised away from their minima (F-11). Beyond the fixtures, the certified regime has to be widened from vacuum with no cutoff to whatever solvent treatment a production run will use, which is a further round of the same term-by-term comparison and not an extrapolation from these numbers. The criterion for judging the study that follows should not be whether the Monte Carlo engine reproduces a molecular dynamics trajectory. It should be whether the engine reaches configurations that molecular dynamics does not reach in comparable time, at energies that mean the same thing in both. The present paper speaks only to the second half of that sentence, and only on the chemistry and the geometries it tested.

5 Availability

MOSAICS Studio. The page is served as static files on the MOSAICS project site at <https://www.cs.ox.ac.uk/mosaics/mosaics-studio/>. It needs a browser with WebAssembly and Web Workers, and WebGL for the molecular viewer, about 25 MB on first visit and a few hundred megabytes of memory during a run; the version described here is 1.2.6. Chrome is the browser the work reported here was run in. On 8 September 2026 the viewer failed to initialise under Safari on one machine while the same page rendered under Chrome, so the browser matrix is stated as tested rather than as supported. The source is at <https://github.com/omagebright/mosaics-studio> under the MIT licence. The page carries the MOSAICS engine compiled to WebAssembly, redistributed with the permission of the engine’s author; the force-field data are distributed as separate works and the underlying parameter sets should be cited to their original publications, which Section 2.2 names.

The engine. MOSAICS [44] is developed at <https://www.cs.ox.ac.uk/mosaics/>. The evaluation code, the deck reader, the converters that write the decks, the reference builders and every single-point input used here live in the engine’s source distribution, with the decks under `params/` and the structures under `examples/`, and the inputs and outputs behind every number in this paper are kept in the evidence tree that accompanies the manuscript sources, one directory per force field and system. That distribution is at present cited by name and project page: it carries no DOI and no public release tag, so a reader cannot yet fetch the exact tree these numbers came from. An archived, versioned deposit with a licence statement is needed and does not yet exist; it is the largest single obstacle to reproducing this paper, and closing it means publishing an artefact, not running anything again.

Manuscript, evidence and data. The manuscript sources, the scripts that draw every figure and table, the two data files behind the tables and both evidence trees are deposited at Zenodo under DOI 10.5281/zenodo.22661461 and developed at <https://github.com/omagebright/mosaics-studio-preprint>, under the CC BY 4.0 licence.

Supporting Information. The MOSAICS single-point settings, the deck sizes, the reference-engine control files, the four leaps presets, the commands that read each engine’s output, the CHARMM36 conversion route, the leaps coordinate edits, and the Start and Run steps of the page.

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Author contributions

F.B.O. converted the parameter sets, built the reference calculations and the evidence tree, wrote MOSAICS Studio and its browser build, and drafted the manuscript. P.M. wrote the MOSAICS engine and the natural-move method, directed the scope and the verification strategy, and revised the manuscript.

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